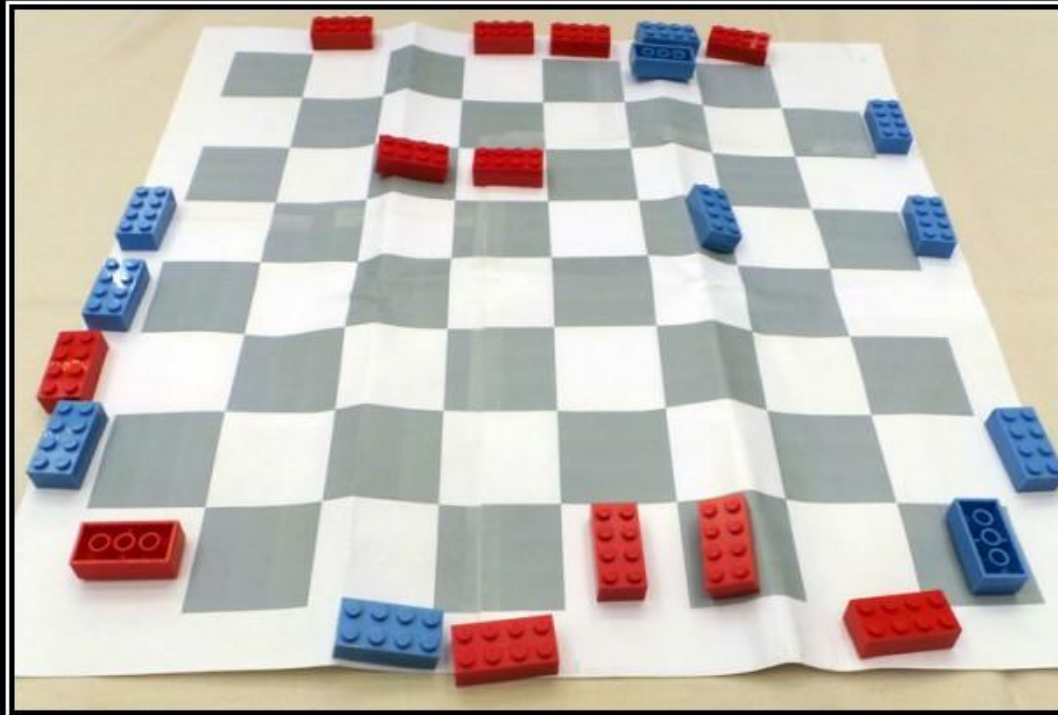




INTRODUCTION TO GAMES

Game Design & Prototyping 01

FOCUS OF THE LECTURE



GAME DESIGN

If you don't make mistakes, you're doing it wrong.
If you don't correct those mistakes, you're doing it really wrong.
If you can't accept that you're mistaken, you're not doing it all.

The background of the slide is a screenshot from the game Hollow Knight. It shows the character Knight standing on a mossy ledge in a dark, cavernous environment. To the right is a large, ornate, spherical structure made of intricate patterns. The scene is dimly lit with greenish-yellow light filtering through the background. In the top left corner, there is a faint overlay of the game's user interface, including a health bar with six heart icons (the first is partially filled) and a double-headed arrow indicating a range or distance.

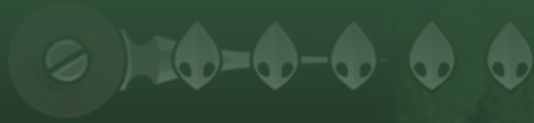
IMPORTANCE OF GAME DESIGN?



<http://stabyourself.net/nottetris2/>



<http://www.youtube.com/watch?v=8edwWVSHsrY>



HI-SCORE
STAB
■ 244 ● 33.4

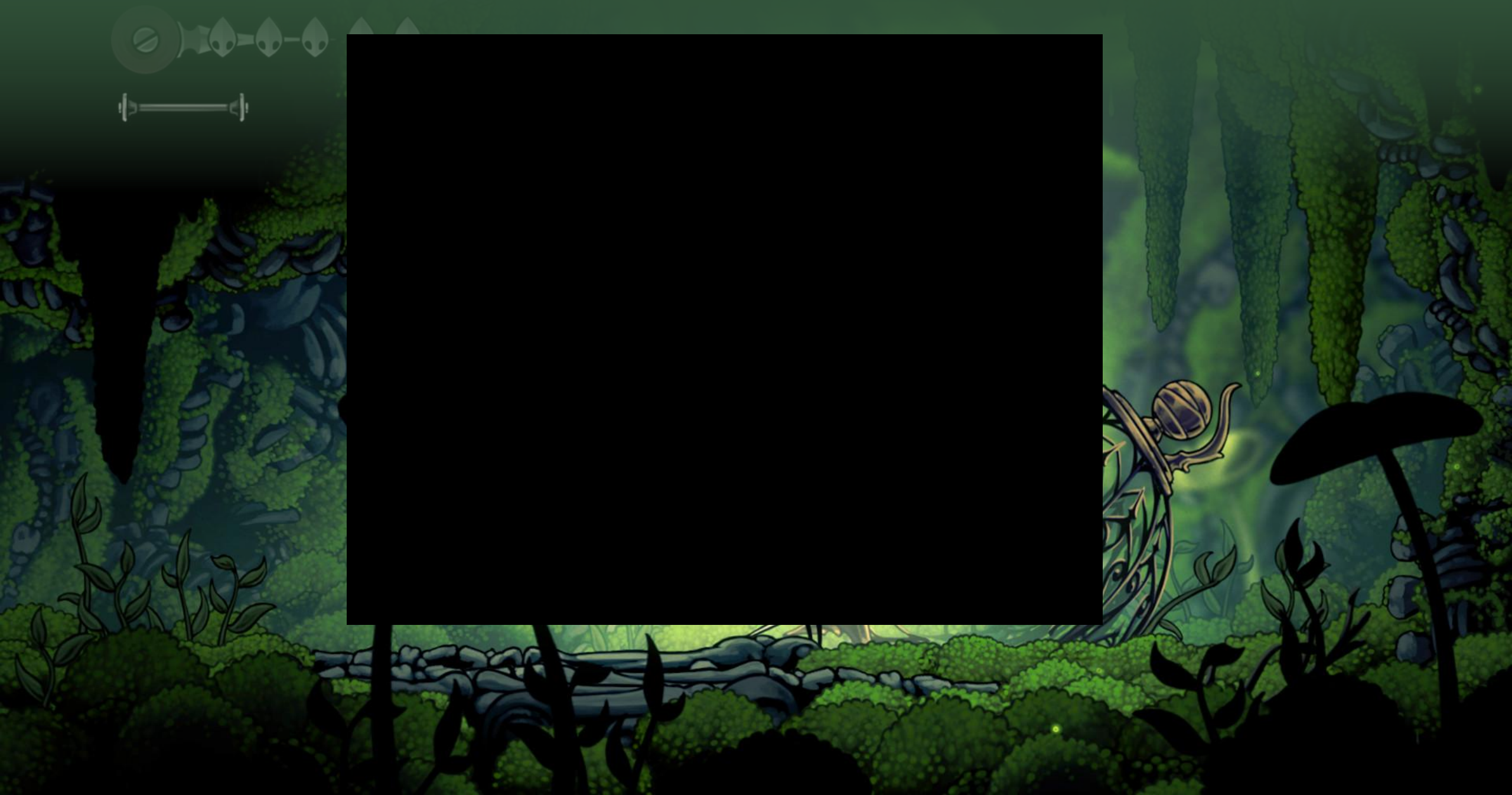
NOT PAC-MAN

▶ PLAY

OPTIONS

maurice™

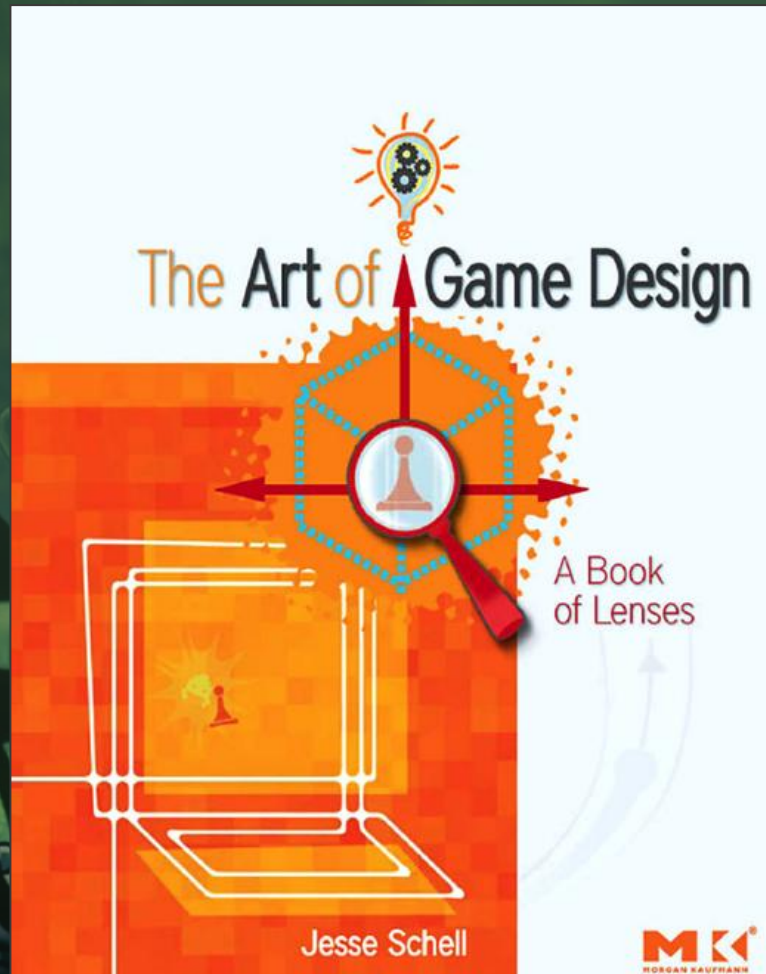
TM & © 2012 2012 MAURICE LTD.
LICENSED TO STABYOURSELF.NET

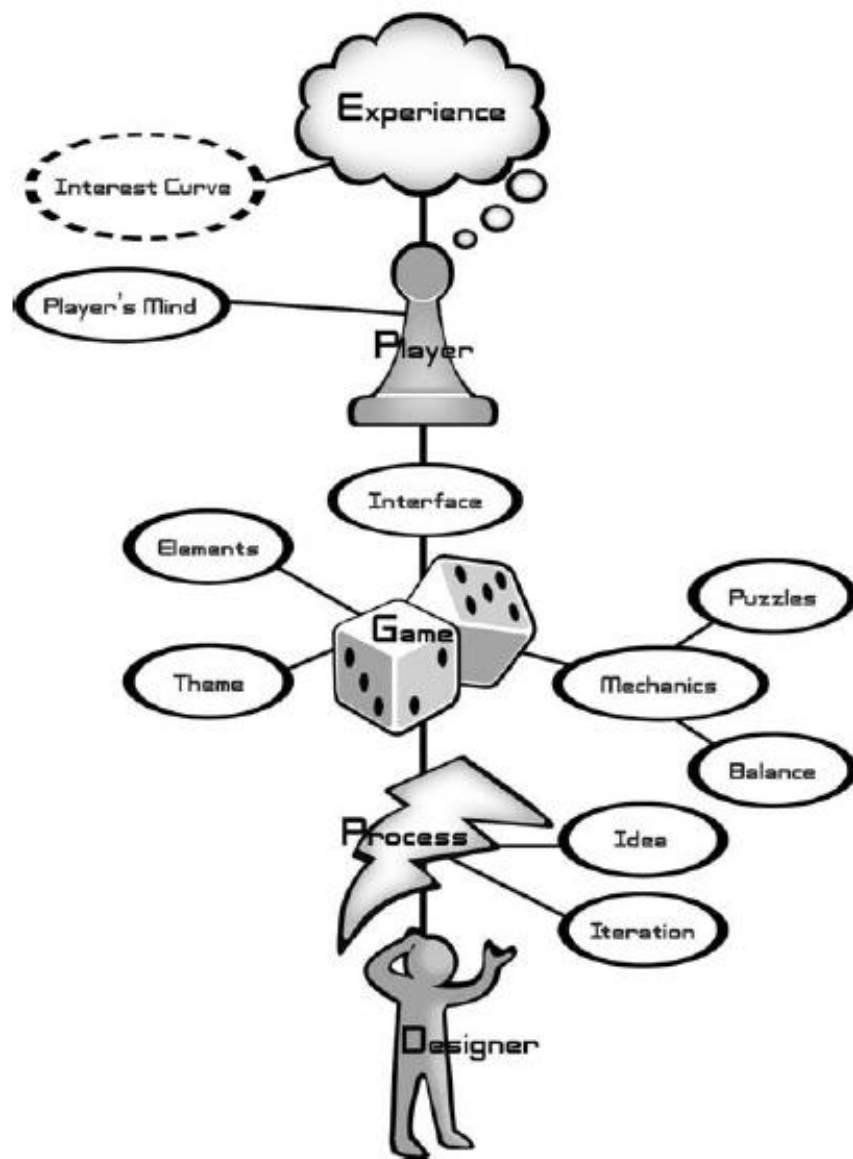


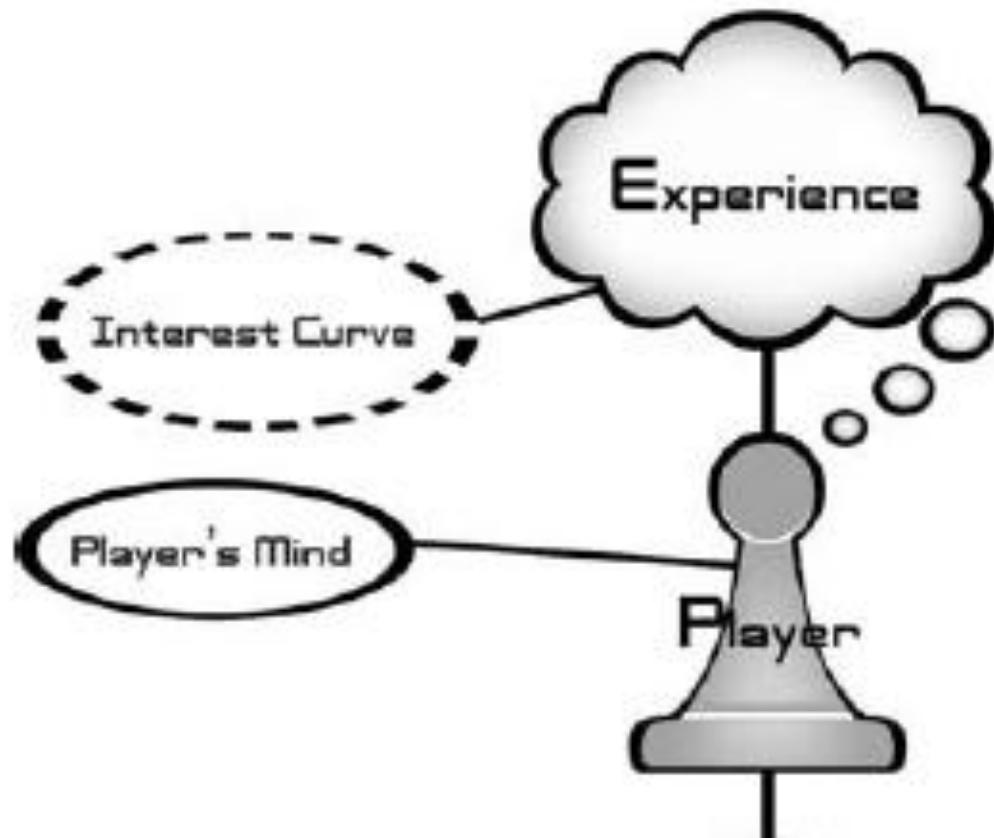
The background of the slide is a screenshot from the game Hollow Knight. In the upper left corner, there is a faint, semi-transparent overlay of the game's user interface, including a health bar with six heart icons and a magic bar with three flask icons. The main scene depicts the character Hollow Knight, a small white insect-like figure with a red skirt, standing on a mossy ground. The knight is facing a large, ornate, spherical structure made of intricate, golden-brown patterns, which appears to be a boss or a significant environmental element. The surrounding environment is a dark, lush forest with large, dark mushrooms and hanging vines, illuminated by soft, greenish light filtering through the trees.

GAME DESIGN: WHERE TO START?

JESSE SCHELL (2008): THE ART OF GAME DESIGN A BOOK OF LENSES









(GAME) EXPERIENCE?



THE LENSE OF ESSENTIAL EXPERIENCE

- Feel mighty?
- Feel smart?
- Feel strong?
- Feel puzzled?





WHAT CONTRIBUTES TO THE EXPERIENCE?



TRANSFER TO GAMES?





EXPERIENCE - EXAMPLE AMNESIA





EXPERIENCE - EXAMPLE AMNESIA





EXPERIENCE - EXAMPLE AMNESIA





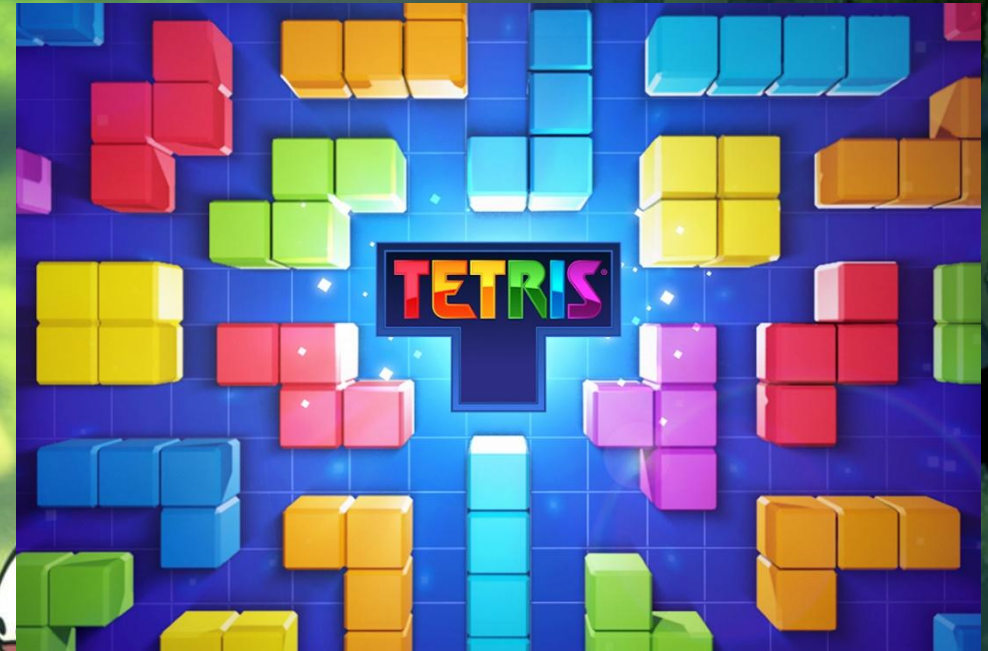
EXPERIENCE - EXAMPLE AMNESIA



The background is a screenshot from the game Hollow Knight. It shows a character with a white mask and a red skirt standing on a mossy ledge in a dark, green, and foggy environment. To the right of the character is a large, ornate, golden structure resembling a nest or a piece of ancient machinery. The scene is dimly lit with greenish-yellow light filtering through the fog. In the top left corner, there are faint UI elements: a health bar with six segments (the first is crossed out) and a double-headed arrow indicating a range or distance.

YOUR (BEST) GAME EXPERIENCE?

STRUCTURE OF AN EXPERIENCE





FACTOR 1: INHERENT INTEREST





FACTOR 2: POETRY OF PRESENTATION



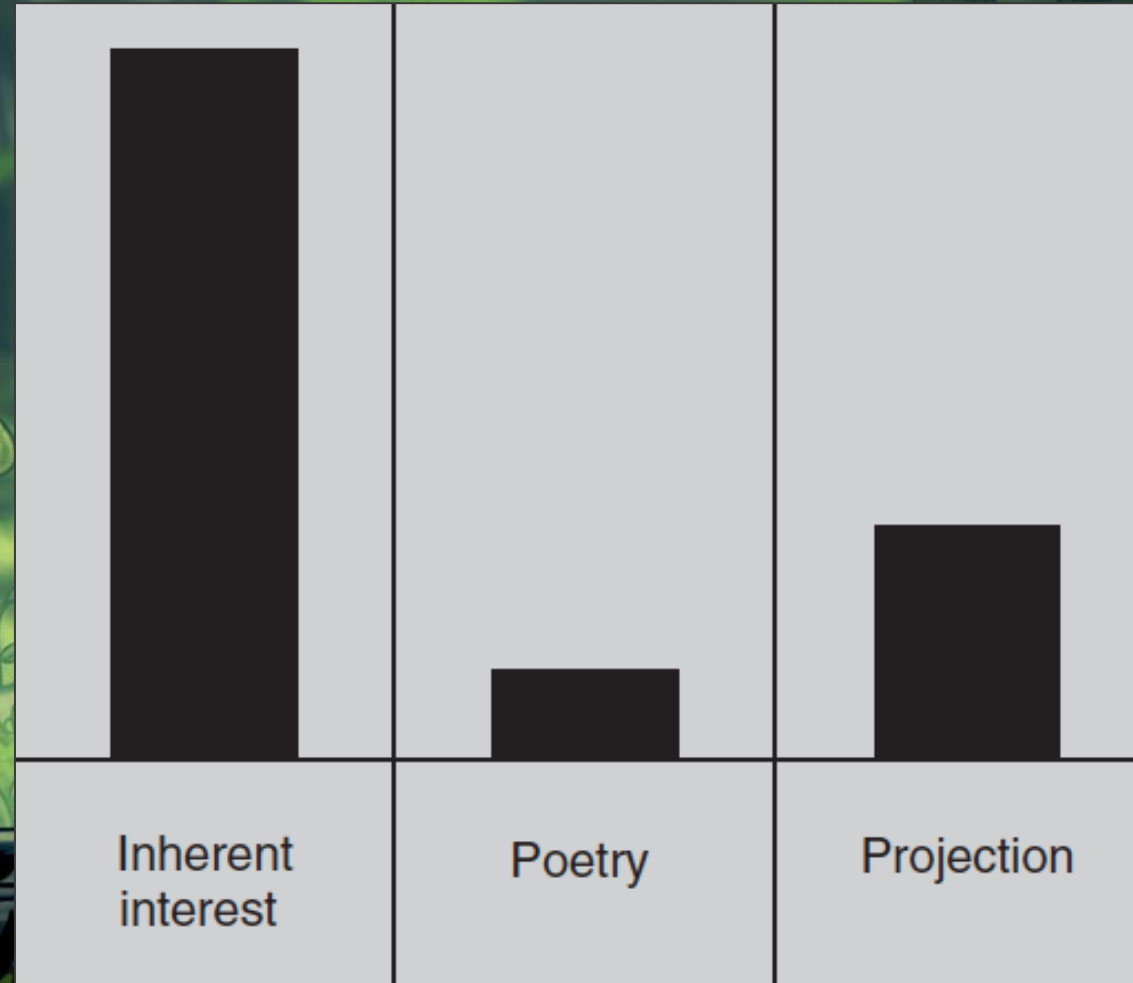


FACTOR 3: PROJECTION



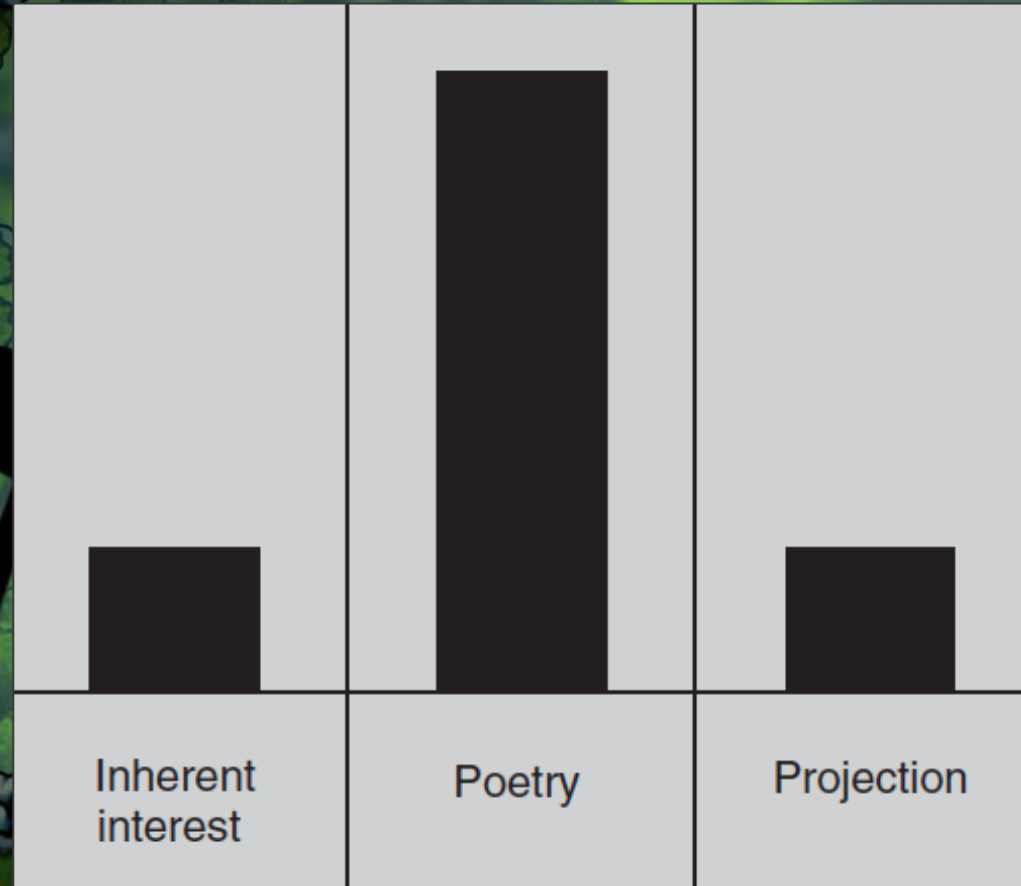


FIRE EATER



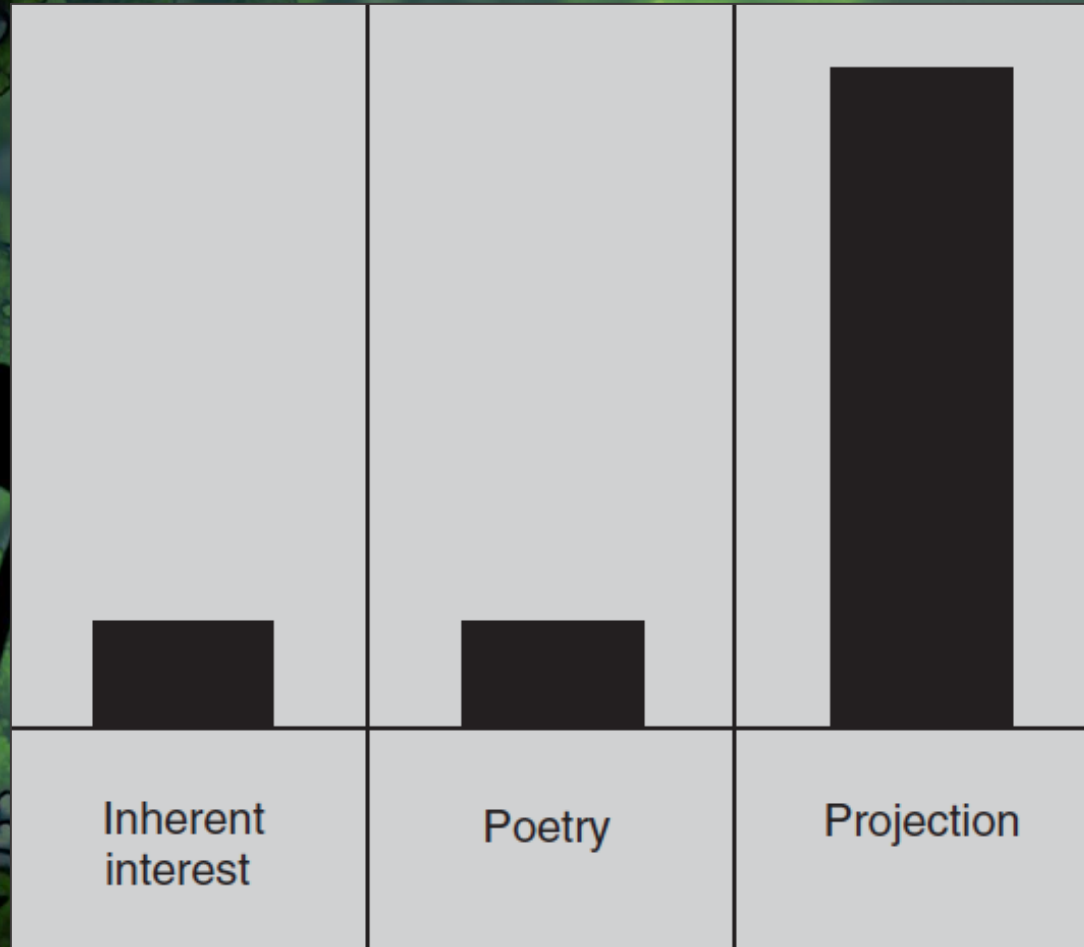


VIOLIN PERFORMANCE





TETRIS



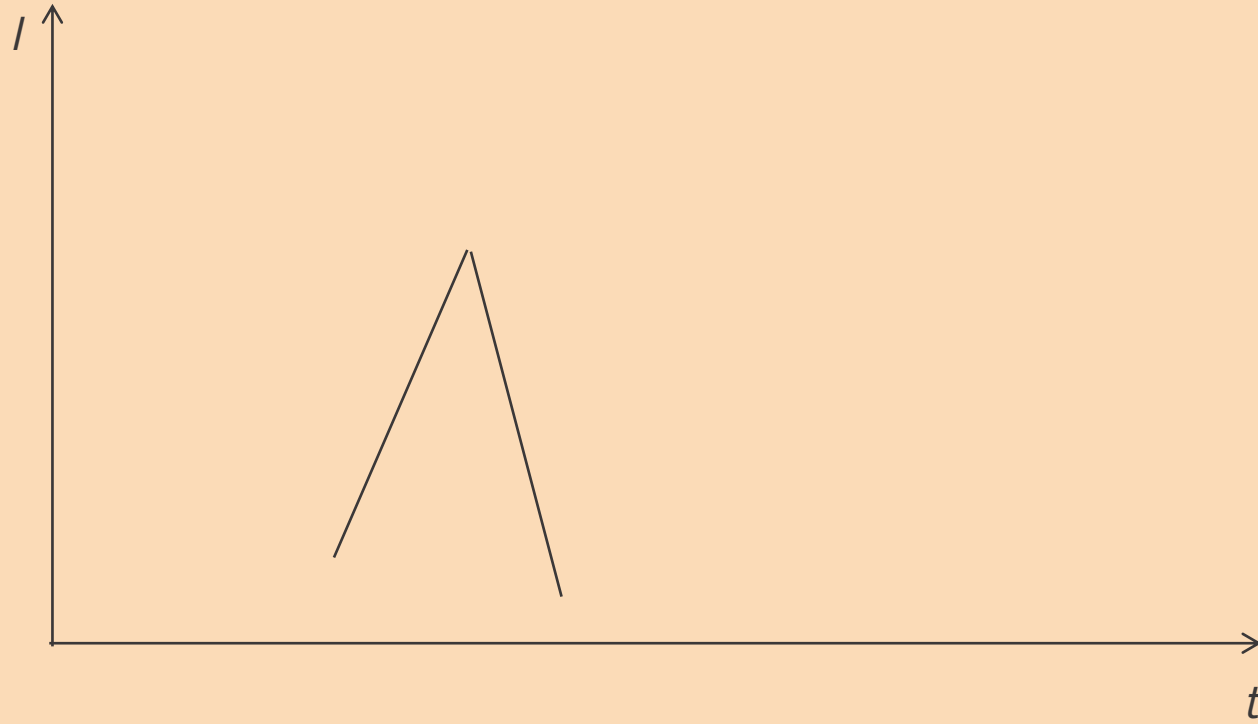


STRUCTURE OF AN EXPERIENCE

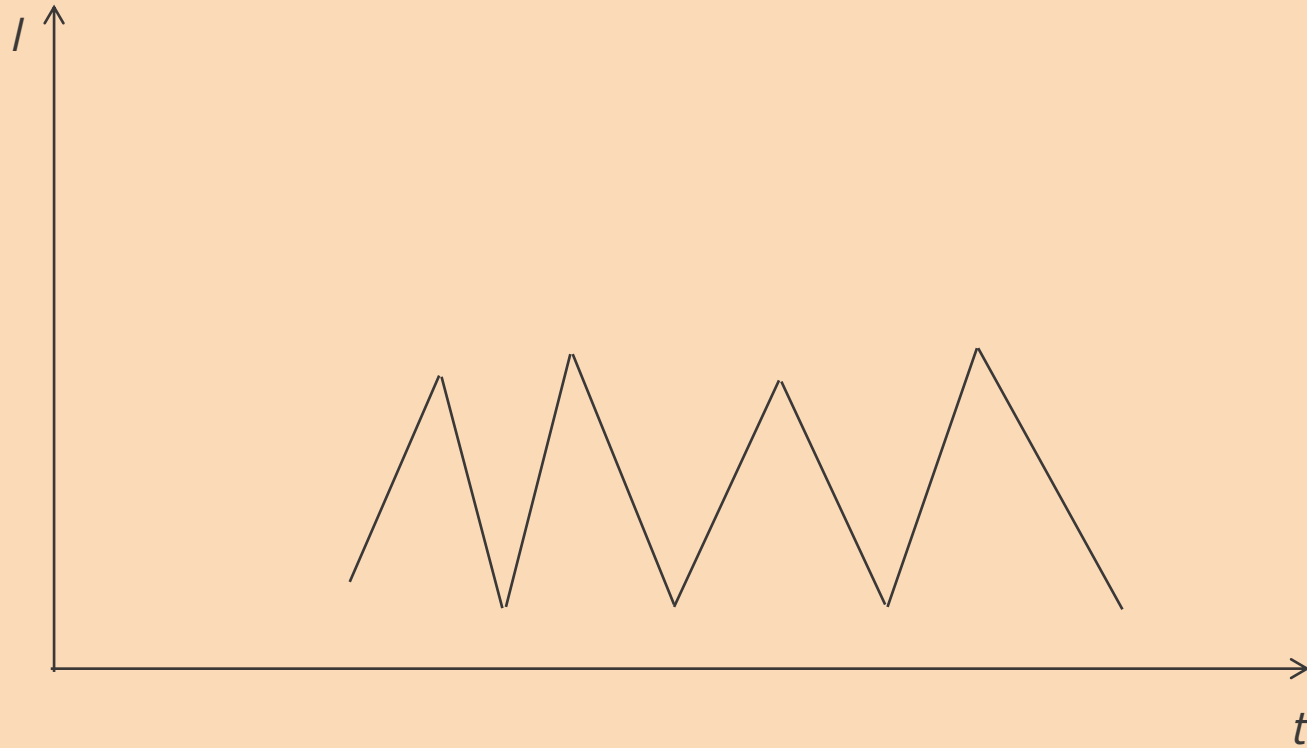




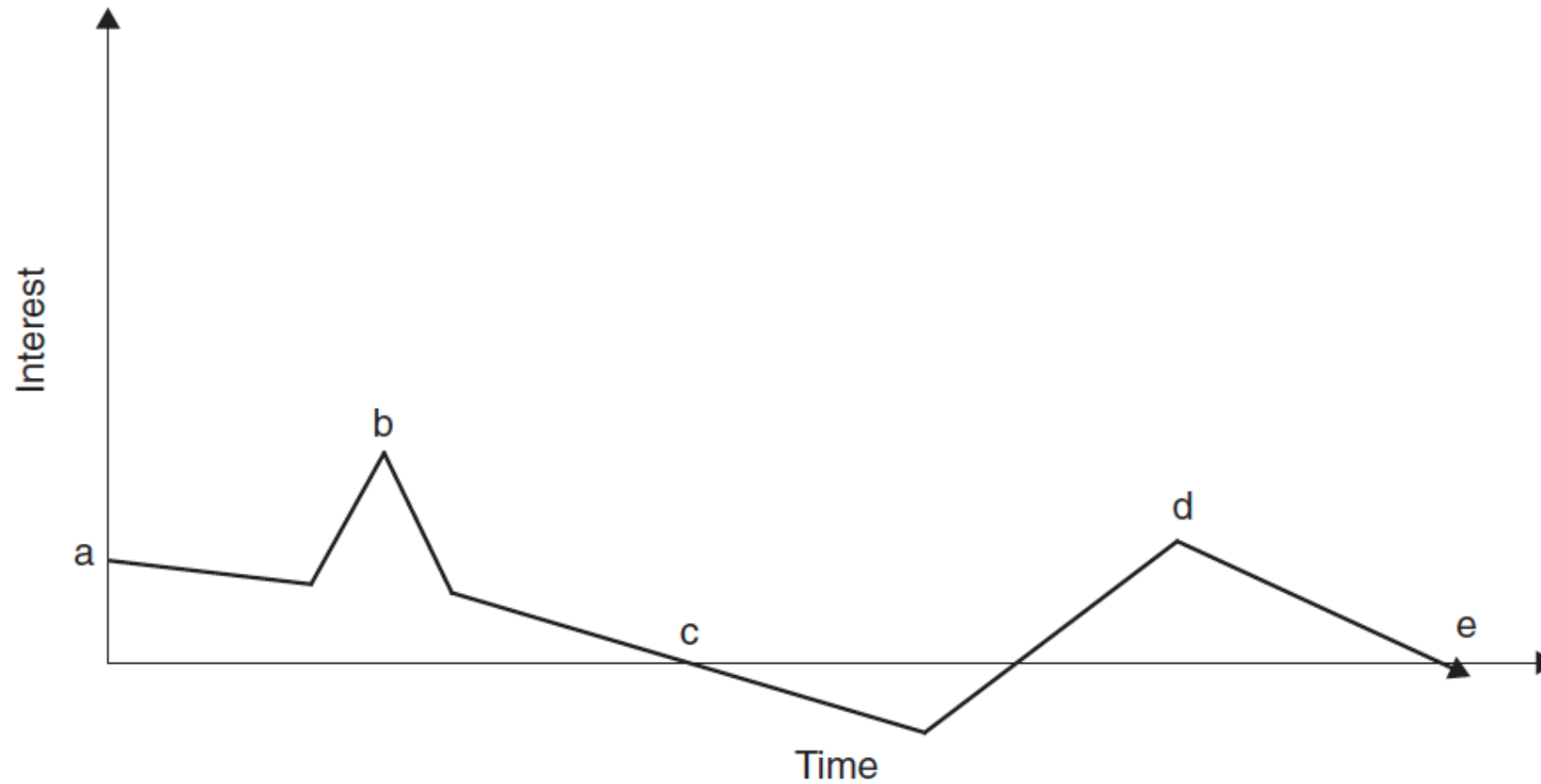
STRUCTURE OF AN EXPERIENCE



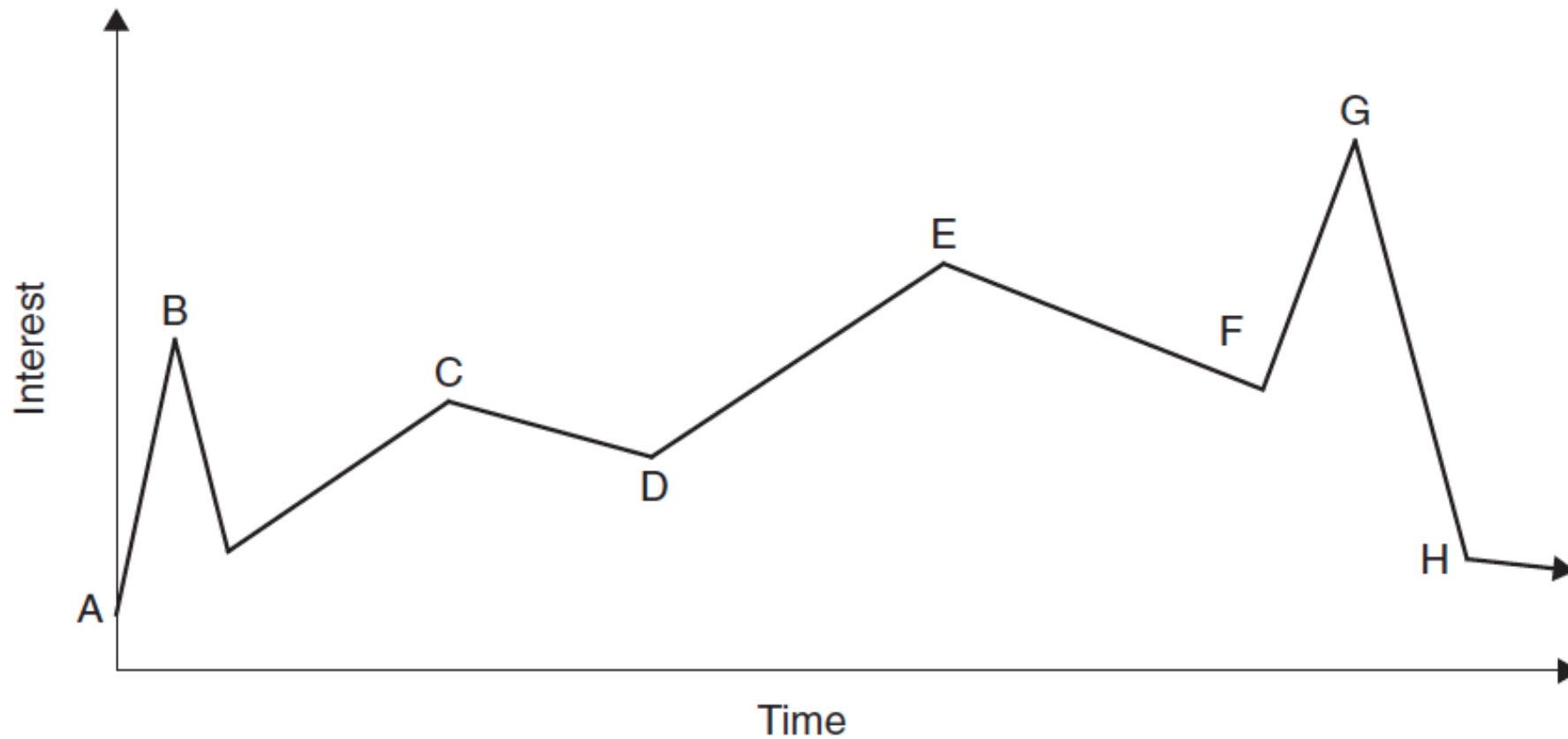
STRUCTURE OF AN EXPERIENCE



STRUCTURE OF AN EXPERIENCE

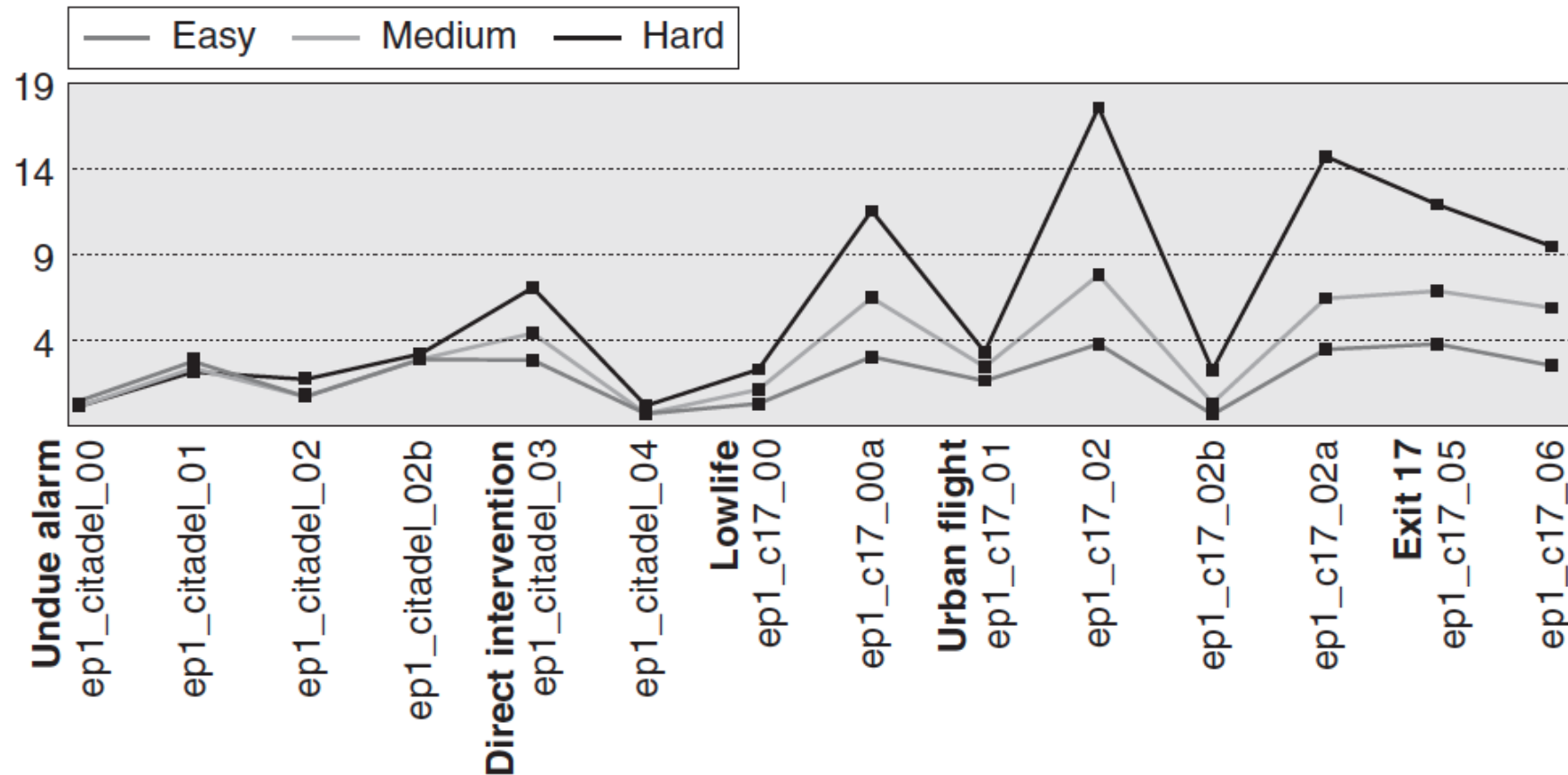


STRUCTURE OF AN EXPERIENCE



STRUCTURE OF AN EXPERIENCE

Average number of deaths (per map)

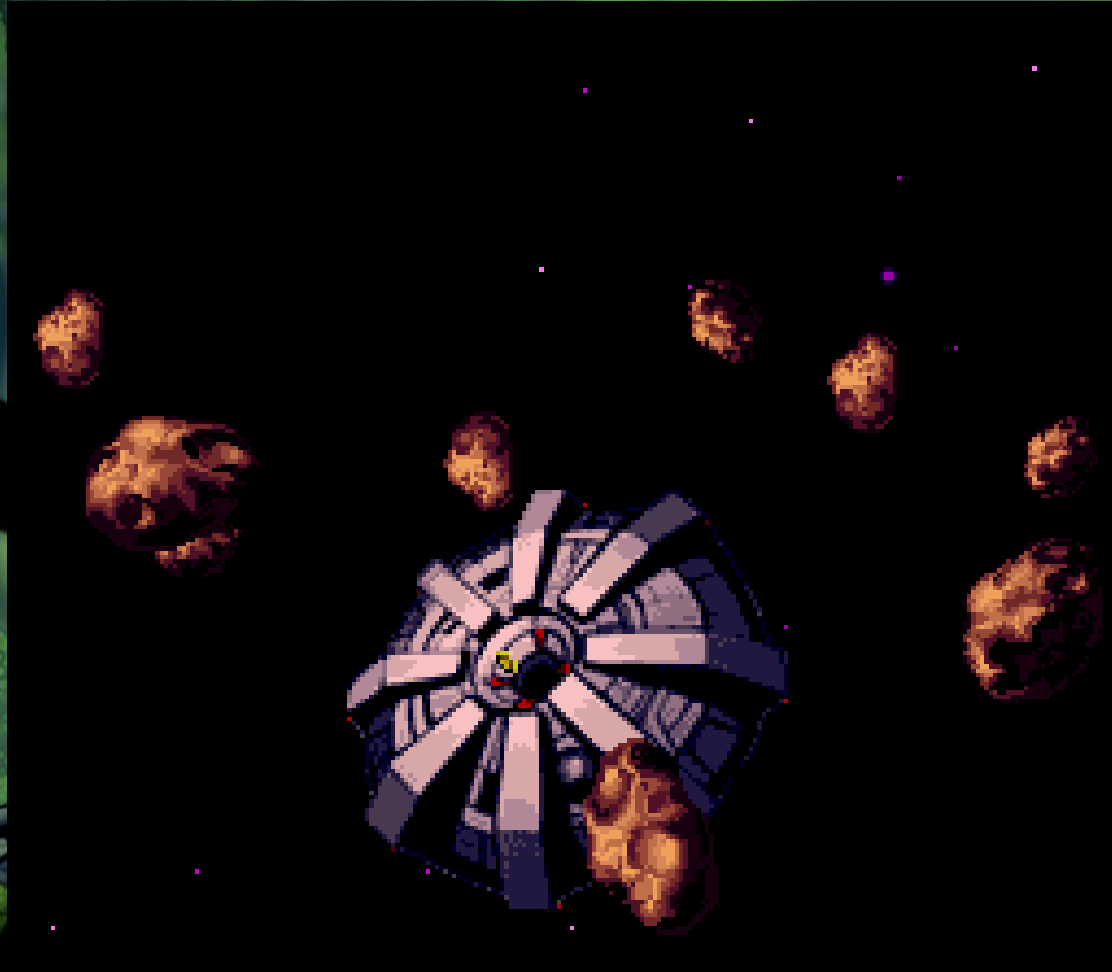


SUPER METROID (1994)





SUPER METROID (1994)





SUPER METROID (1994)





SUPER METROID (1994)





SUPER METROID (1994)





SUPER METROID (1994)

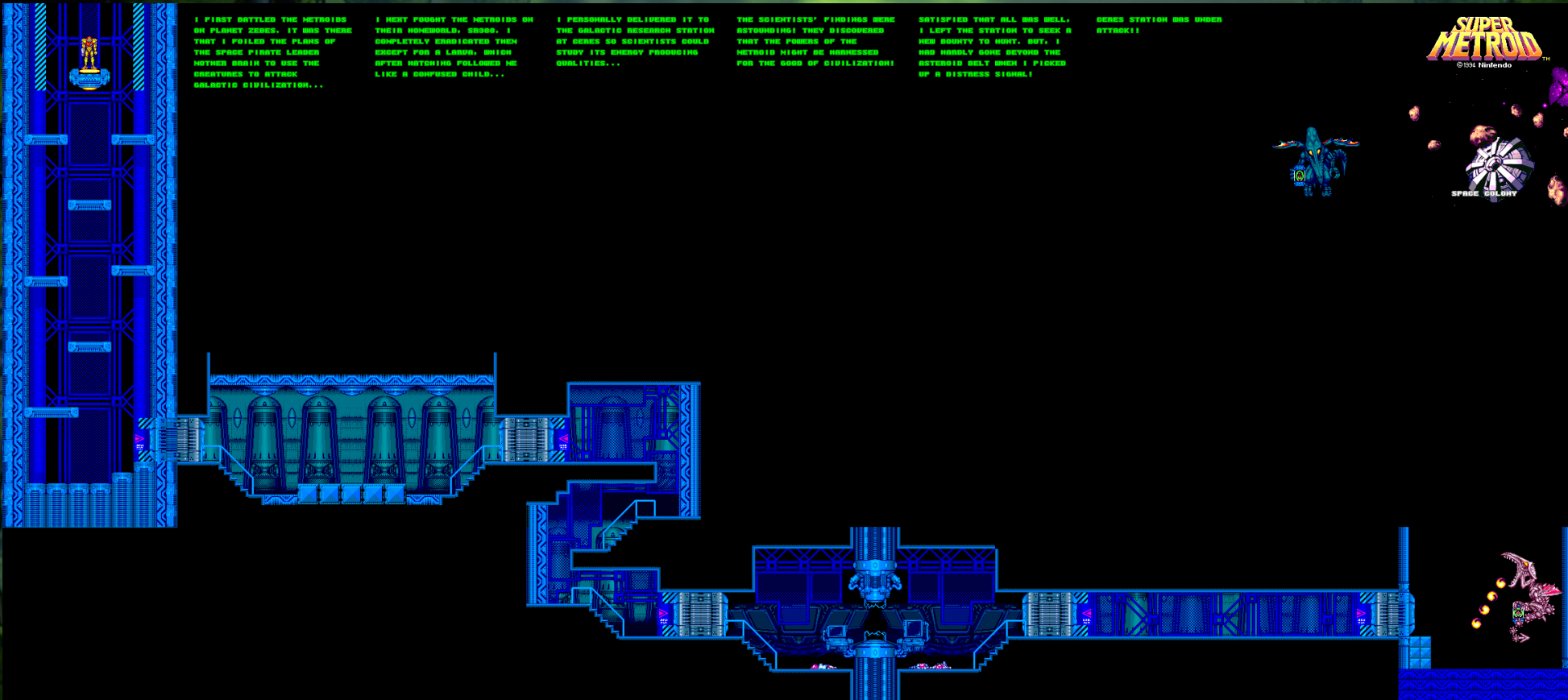


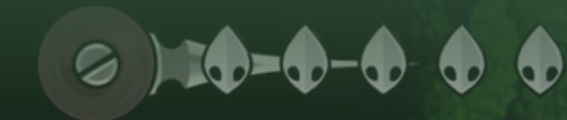


SUPER METROID (1994)

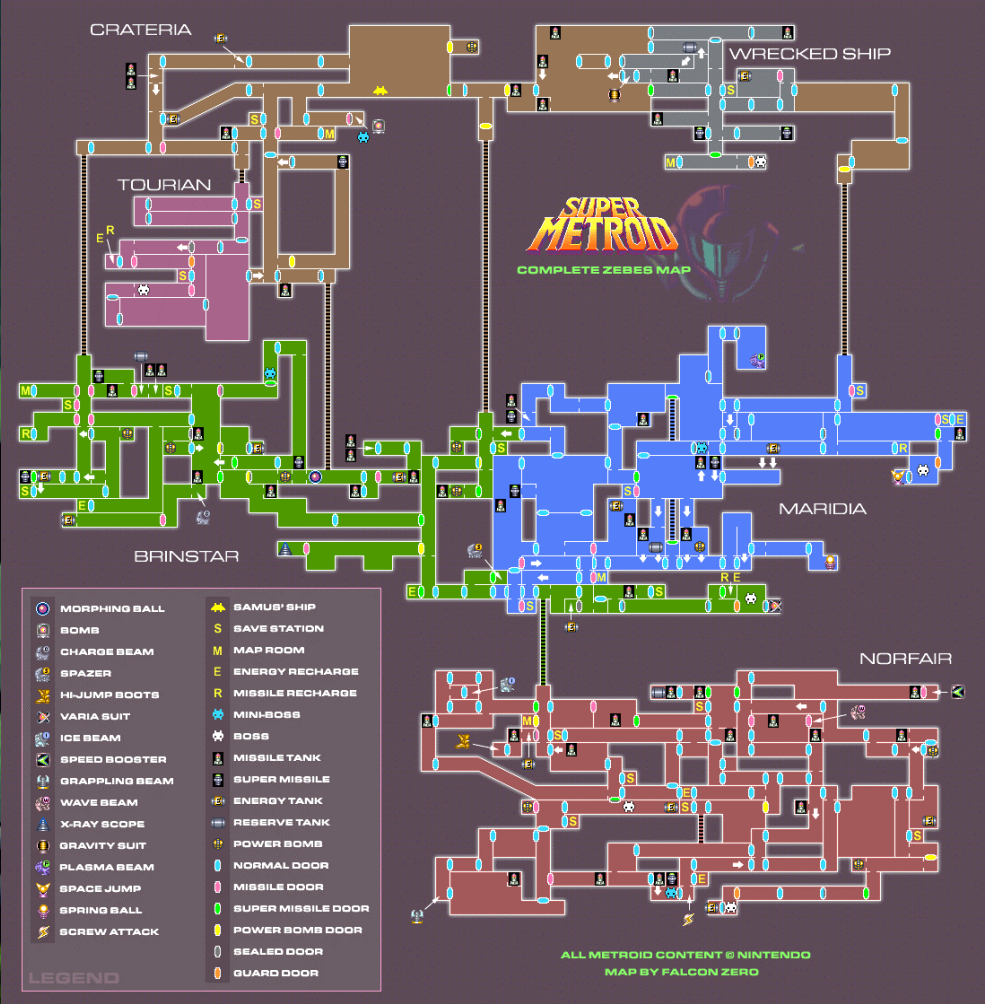


SUPER METROID (1994)



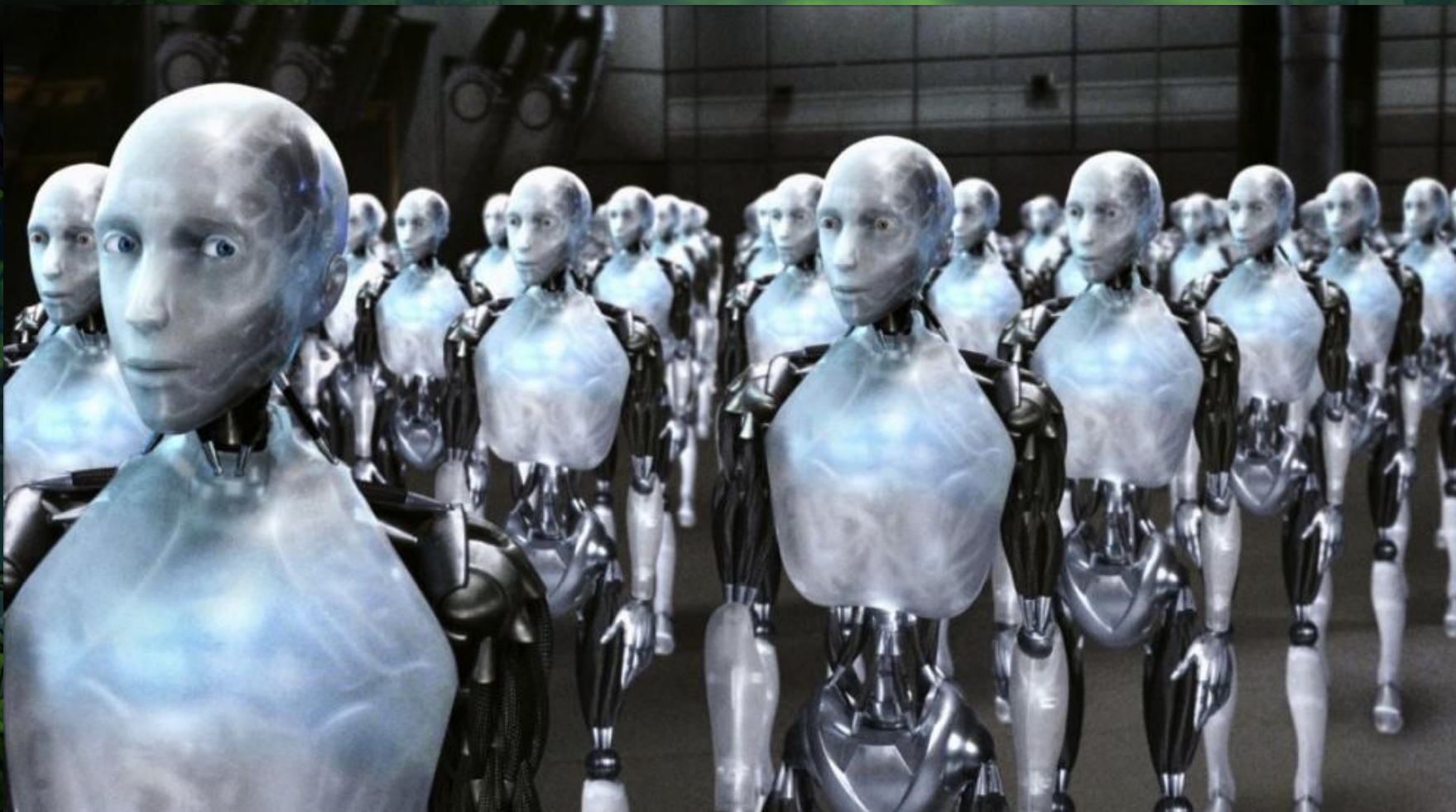


SUPER METROID (1994)



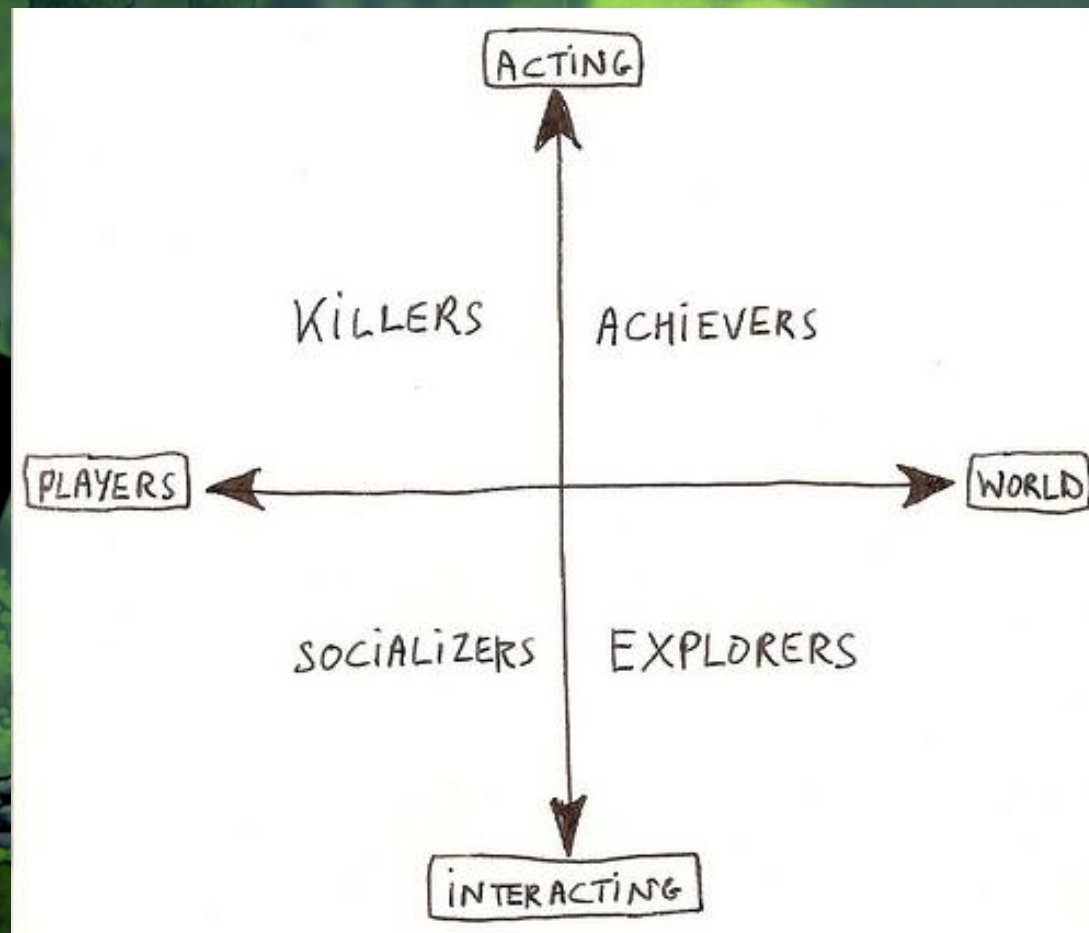


PLAYER TYPES



BARTLE'S PLAYER TYPOLOGY

- Achiever
- Explorer
- Socializer
- Killer



BARTLE'S PLAYER TYPOLOGY

Problems – why not use it?

- Model was based on intuition
- Not been validated with empirical data
- Not been shown that the four types are independent





SEEKER



Journey

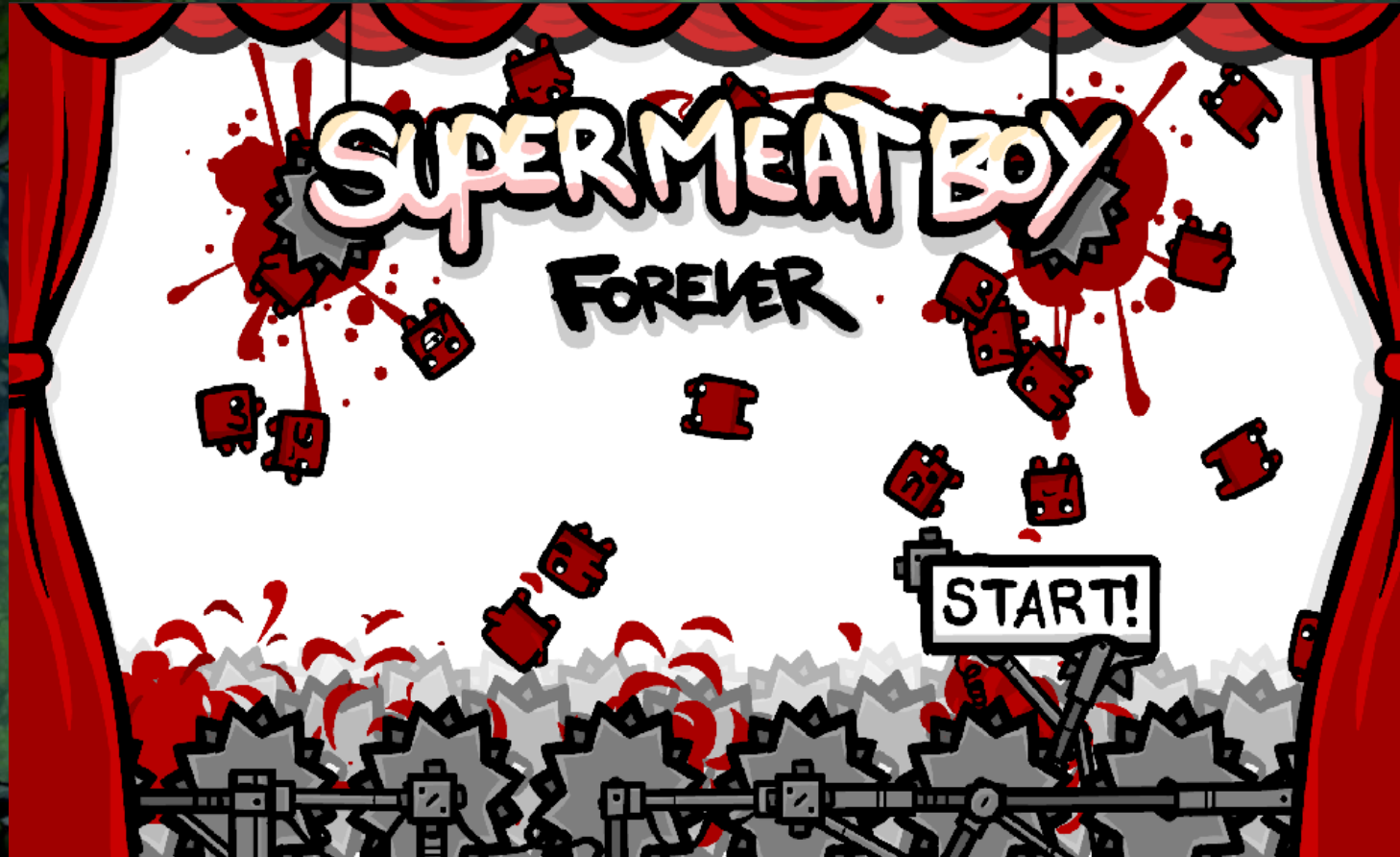


SURVIVOR



Resident Evil

DAREDEVIL



Super Meat Boy



MASTERMIND



The Room



CONQUEROR



ICGAMERS

Unreal Tournament

SOCIALISER



World of Warcraft



ACHIEVER GAME



Angry Birds: Rio



FIND OUT WHO YOU ARE @

<http://survey.ihobo.com/BrainHex/>

The image is a screenshot from the video game Hollow Knight. In the upper left corner, there is a semi-transparent UI overlay showing a health bar with six heart icons (the first is partially filled) and a long, thin stamina bar below it. The main scene is a lush, dark green underground environment. In the center, a small white character with a red skirt and a crescent-shaped head (the Knight) stands on a mossy ledge, looking towards a large, ornate, spherical structure made of intricate, glowing green patterns. The background is filled with dense foliage, hanging vines, and soft, ethereal green light filtering through the space.

WHAT EFFECTS AN EXPERIENCE?

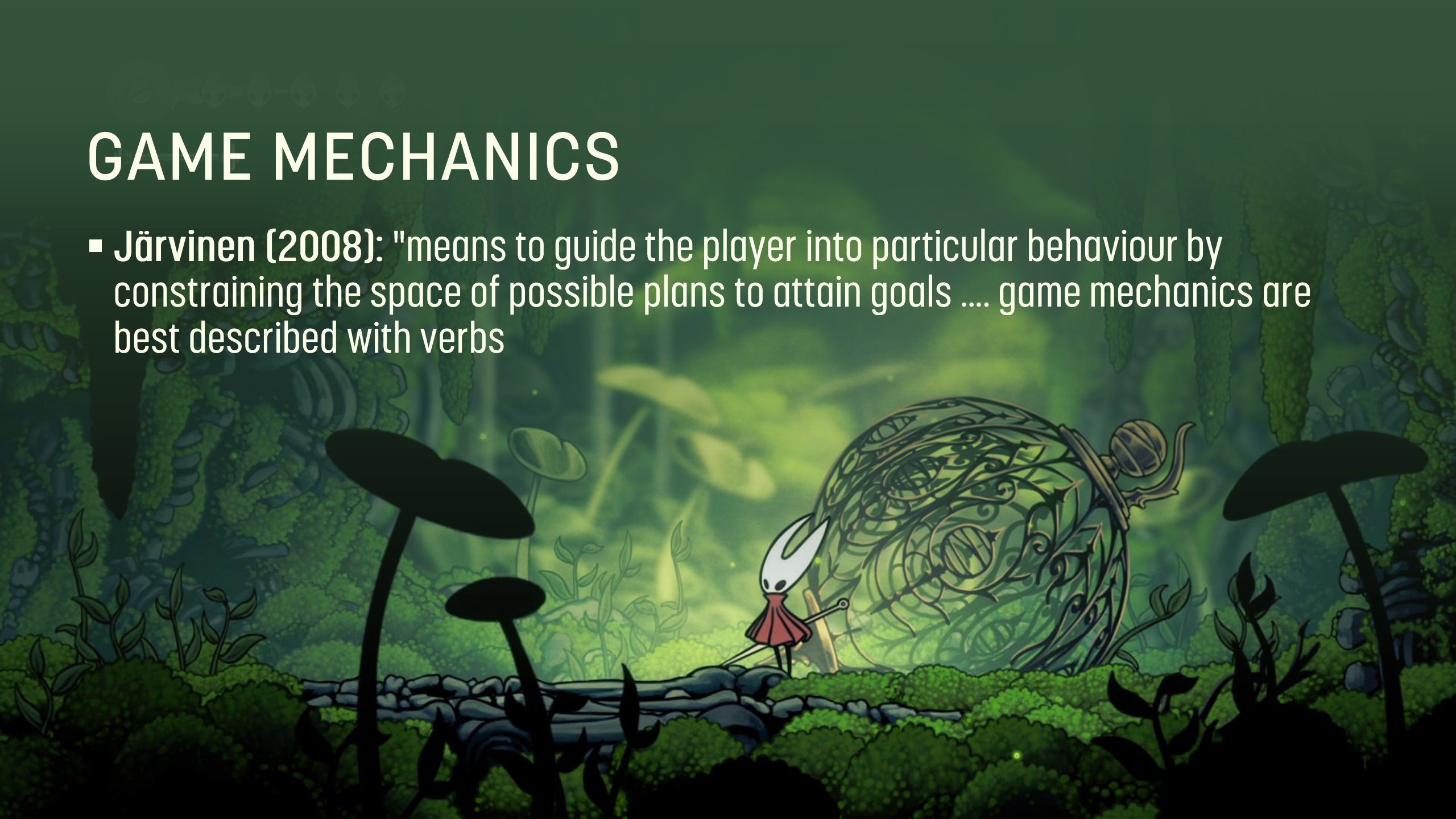


The image shows a screenshot from the game Hollow Knight. In the top left corner, there is a UI element consisting of a row of six icons: a cross, a heart, and four identical leaf-like symbols. Below these is a horizontal bar with a central crosshair. The main scene is a dark, mossy underground area. A small, white, insect-like character with a red skirt and a crescent-shaped head is standing on a stone path. The character is facing a large, spherical, ornate structure made of intricate, golden-brown patterns. The background is filled with dark green moss, hanging stalactites, and large, dark mushrooms. Sunlight filters through the background, creating a hazy, greenish glow.

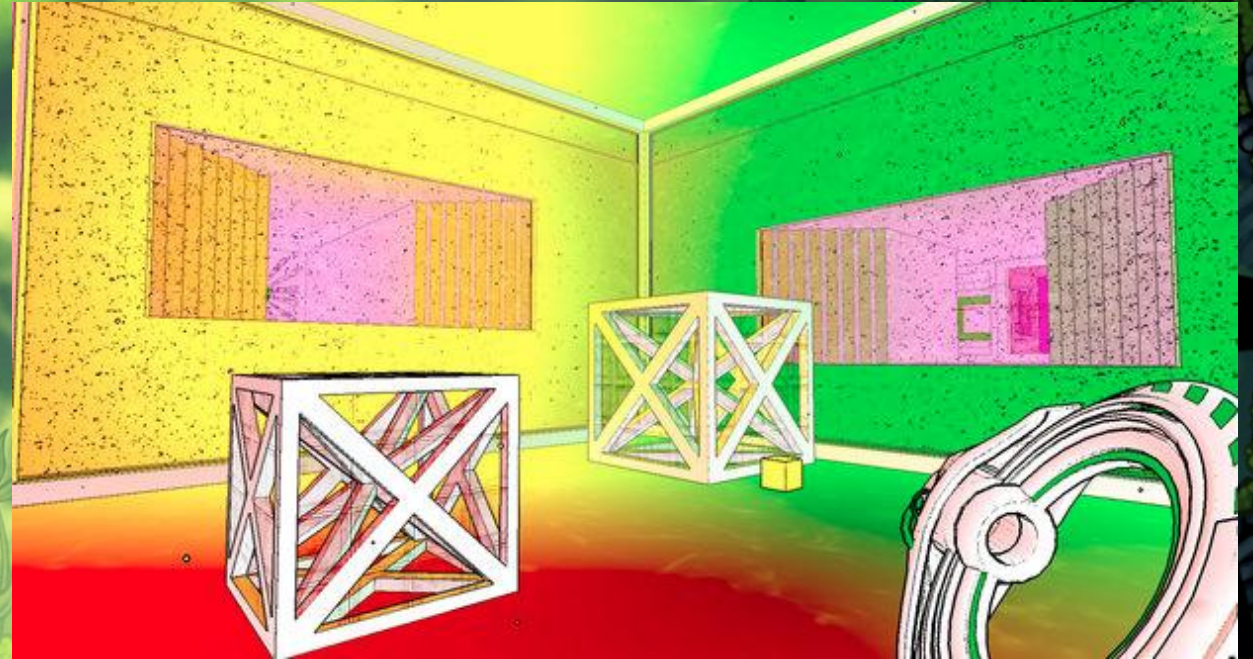
ONE ELEMENT: GAME MECHANICS

GAME MECHANICS

- Järvinen (2008): "means to guide the player into particular behaviour by constraining the space of possible plans to attain goals game mechanics are best described with verbs"



MECHANIC 1: GAME SPACE



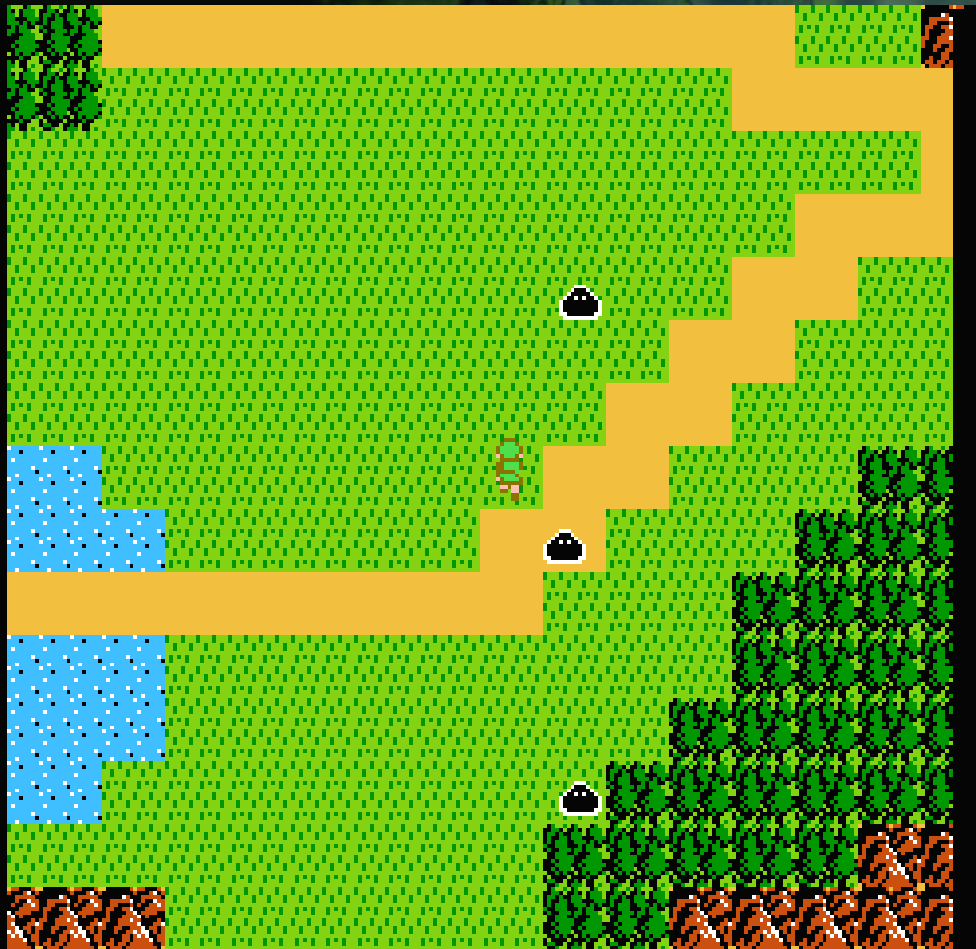
GAME SPACE – DISCRETE OR CONTINUOUS



GAME SPACE – MULTIPLE DIMENSIONS



GAME SPACE – MULTIPLE DIMENSIONS



GAME SPACE – MULTIPLE DIMENSIONS

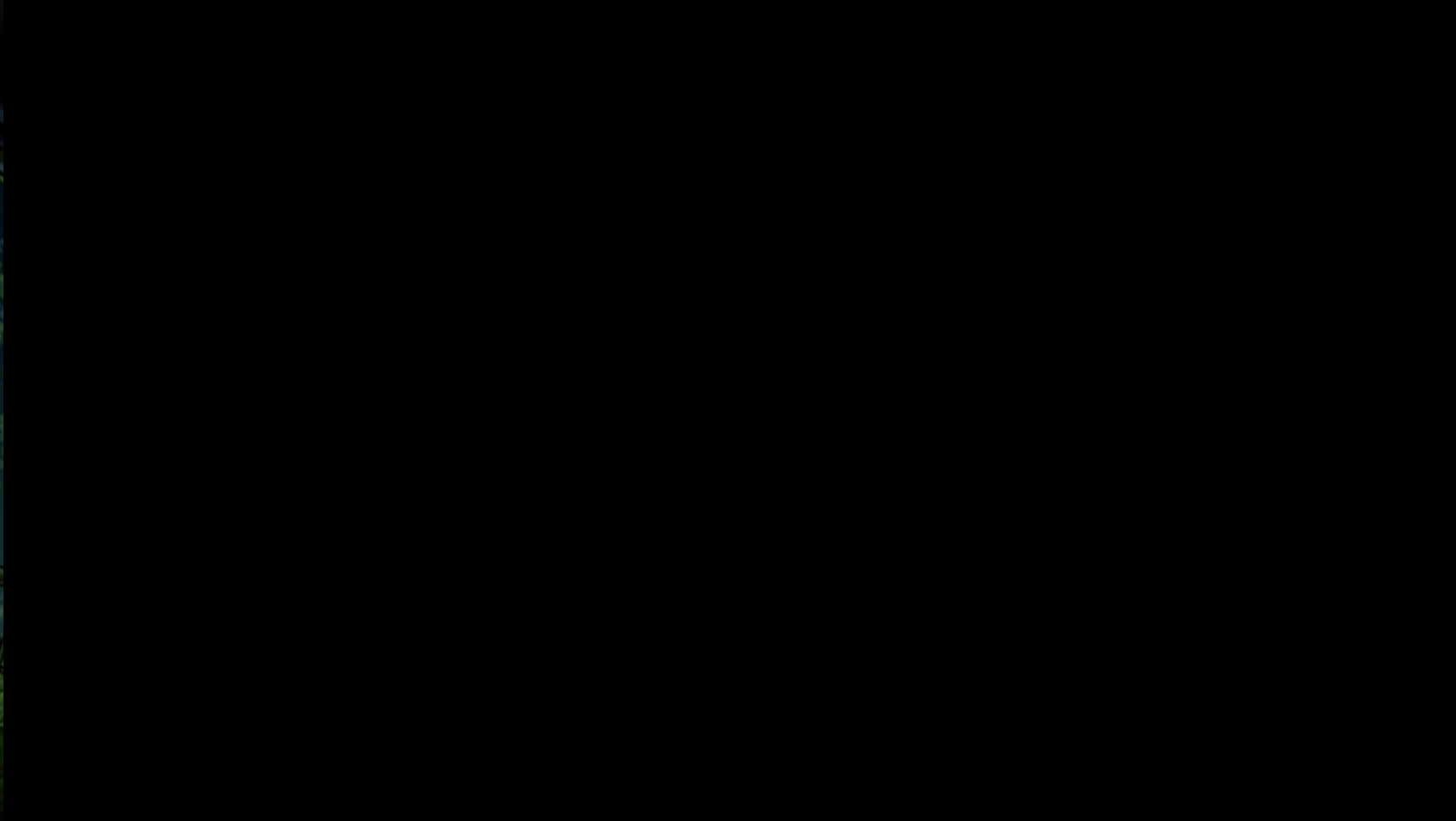


GAME SPACE - ANTICHAMBER





GAME SPACE – ANTICHAMBER



OBJECTS, ATTRIBUTES & STATUS

- Objects
- Characters
- Props
- Point Stats



OBJECTS = NOUNS

Luigi =



ATTRIBUTES = ADJECTIVES



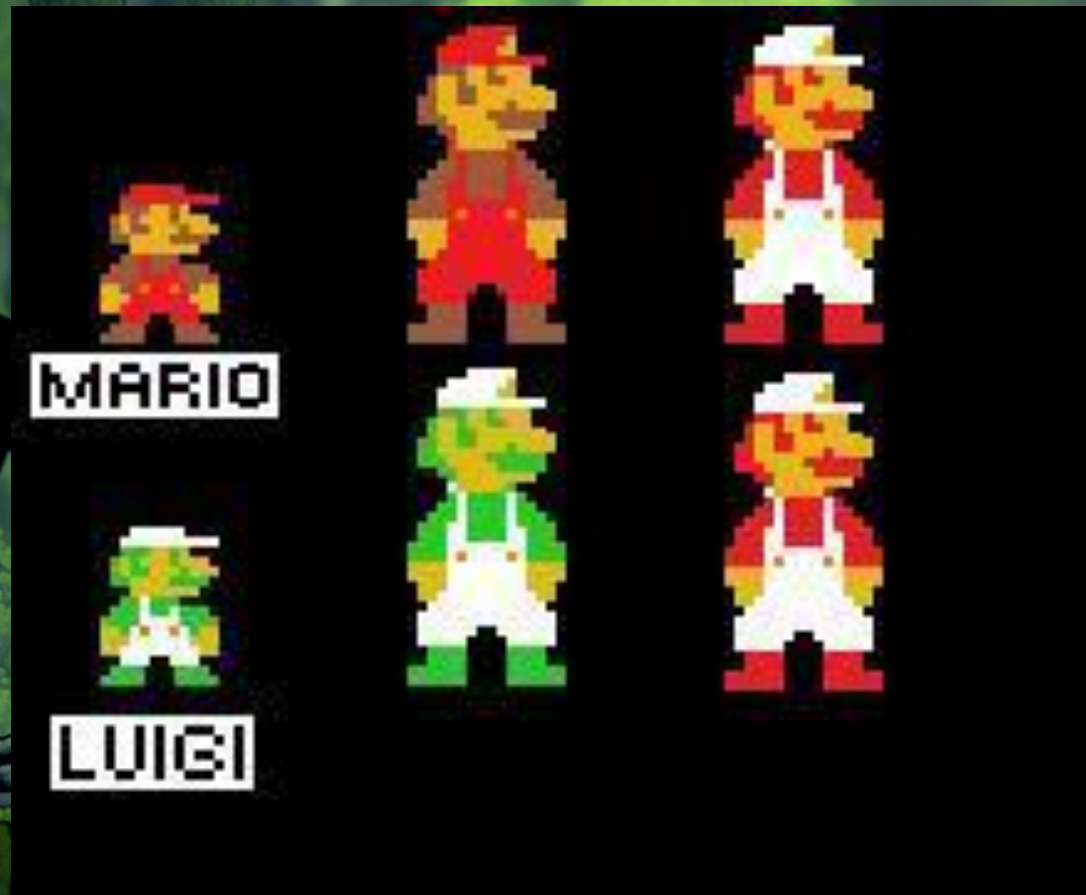
ATTRIBUTES = ADJECTIVES



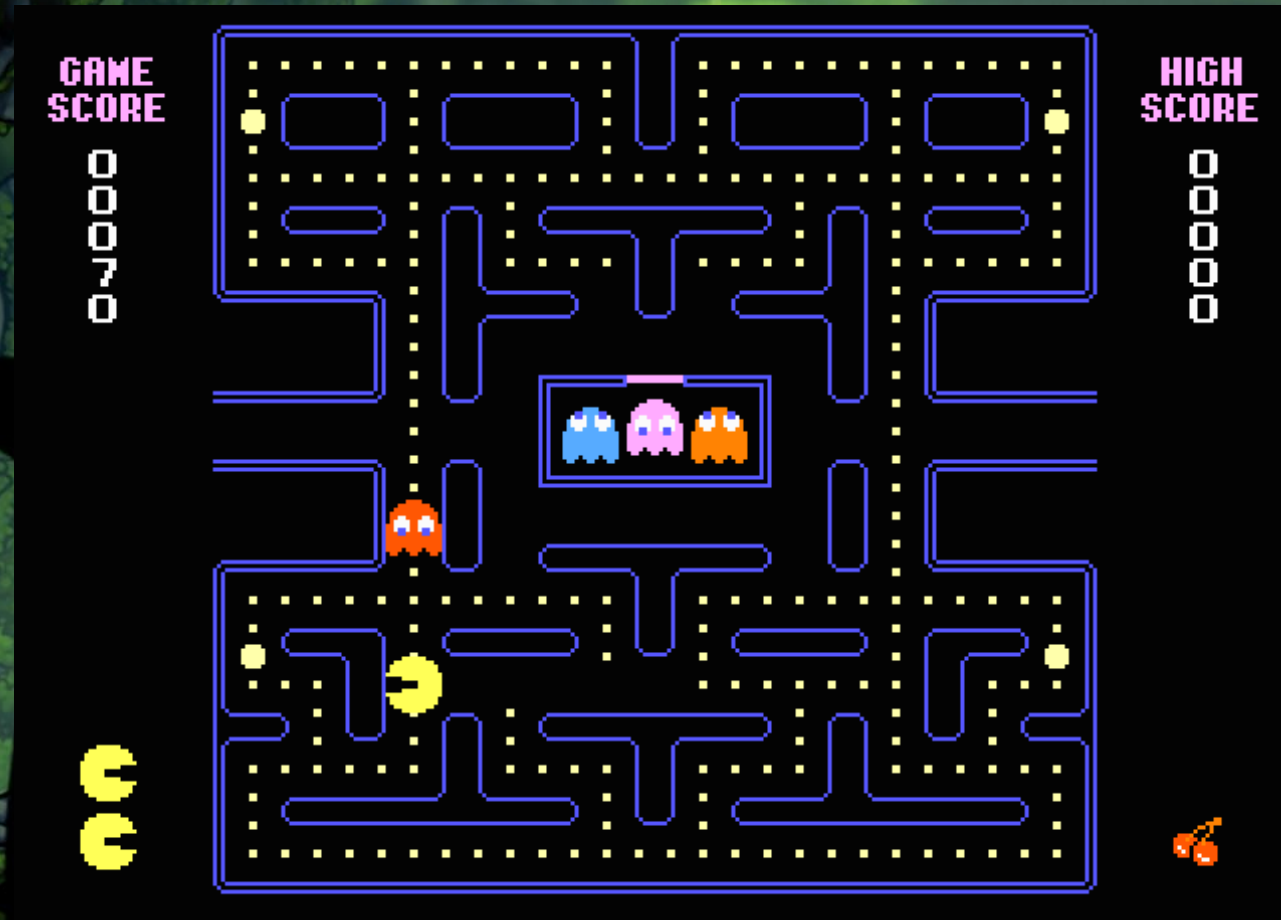
ATTRIBUTES HAVE STATES



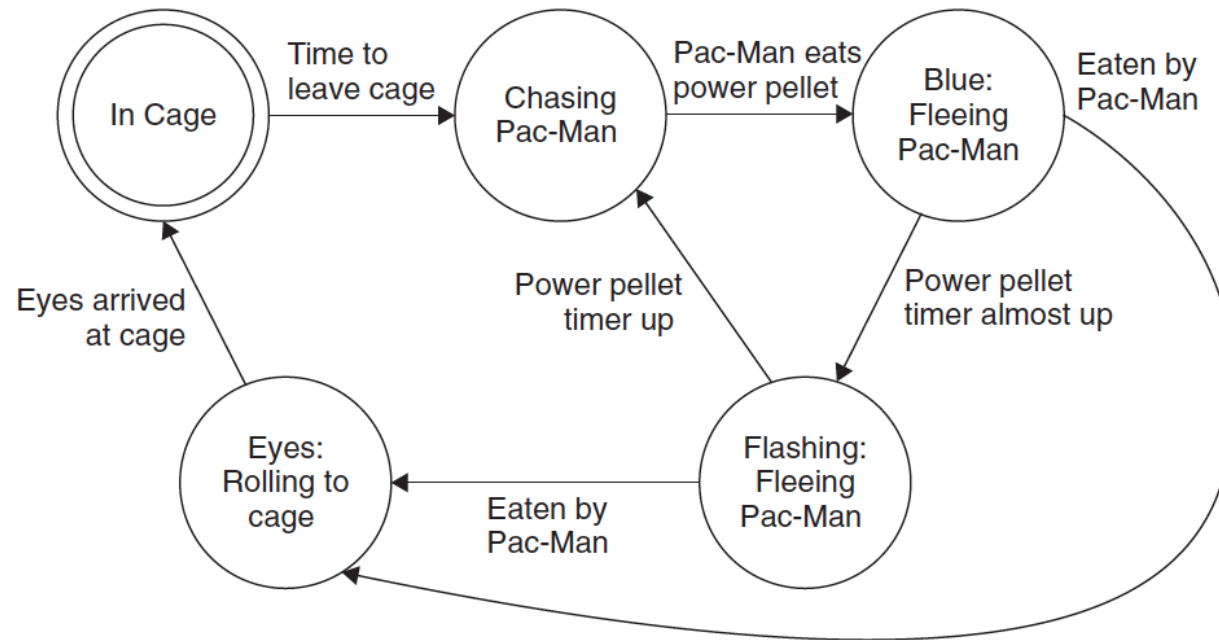
STATES: DYNAMICS OR STATIC



EXAMPLE - PACMAN



EXAMPLE - PACMAN





ARE GAME MECHANICS RELEVANT?



THE SHORT ANSWER...



The background of the slide is a screenshot from the game Goat Simulator. It shows a lush, green, and somewhat chaotic environment with various plants, rocks, and a large, dark, mushroom-like structure in the foreground. In the top left corner, there are UI elements from the game, including a circular icon with a diagonal line and a series of small, stylized goat heads.

THE SHORT ANSWER... GOAT SIMULATOR

<https://www.youtube.com/watch?v=dvWGLcdl8o8>

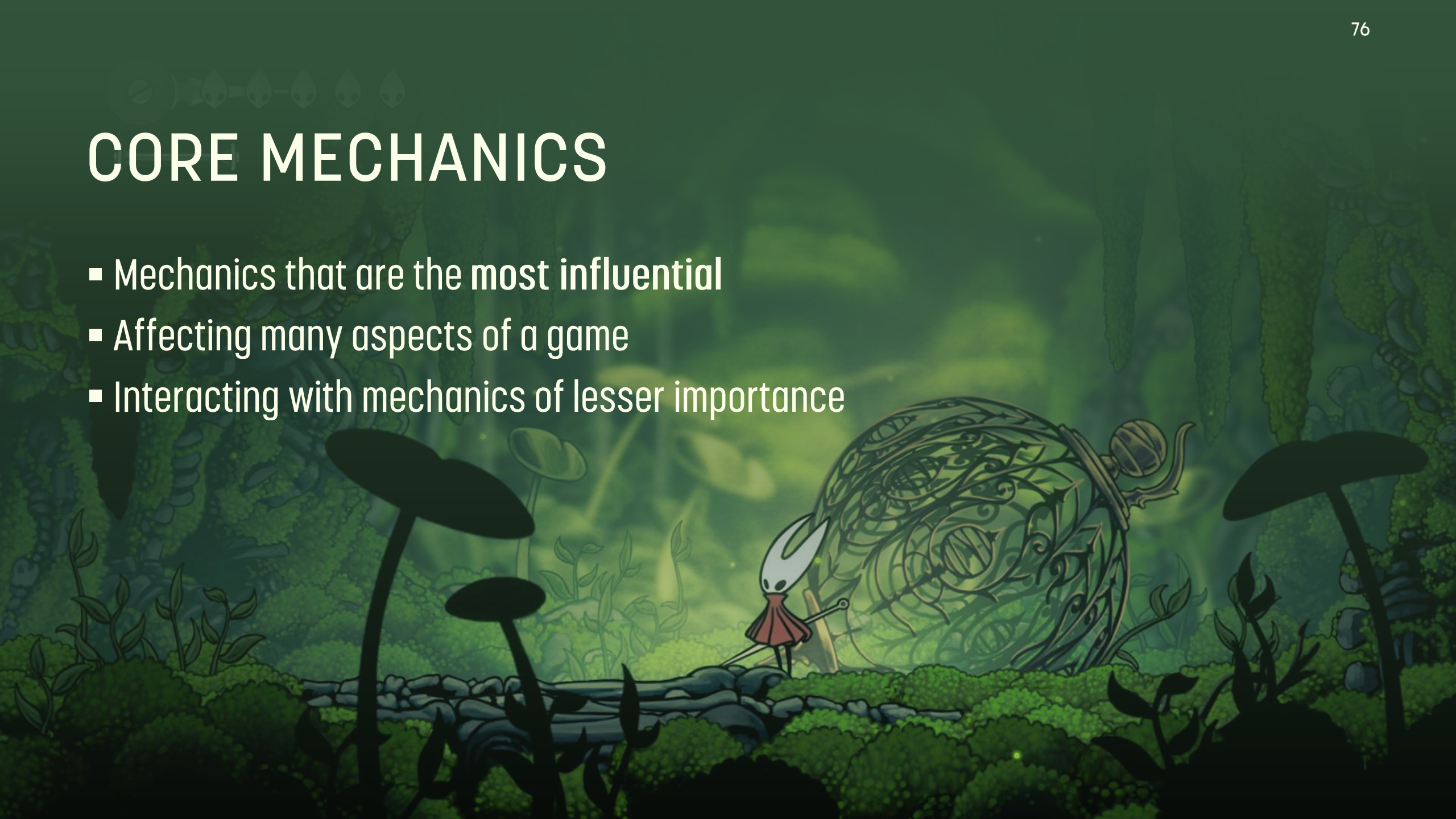
BOOK: GAME MECHANICS

- Title: Game Mechanics: Advanced Game Design
- Ernest Adams & Joris Dormans (2012)



CORE MECHANICS

- Mechanics that are the most influential
- Affecting many aspects of a game
- Interacting with mechanics of lesser importance



CORE MECHANICS

- Example: mechanics that implement gravity in a platform game
- Affect almost all moving objects in the game
- Interact with mechanics for jumping...
- ...or the mechanics that control damage to falling characters



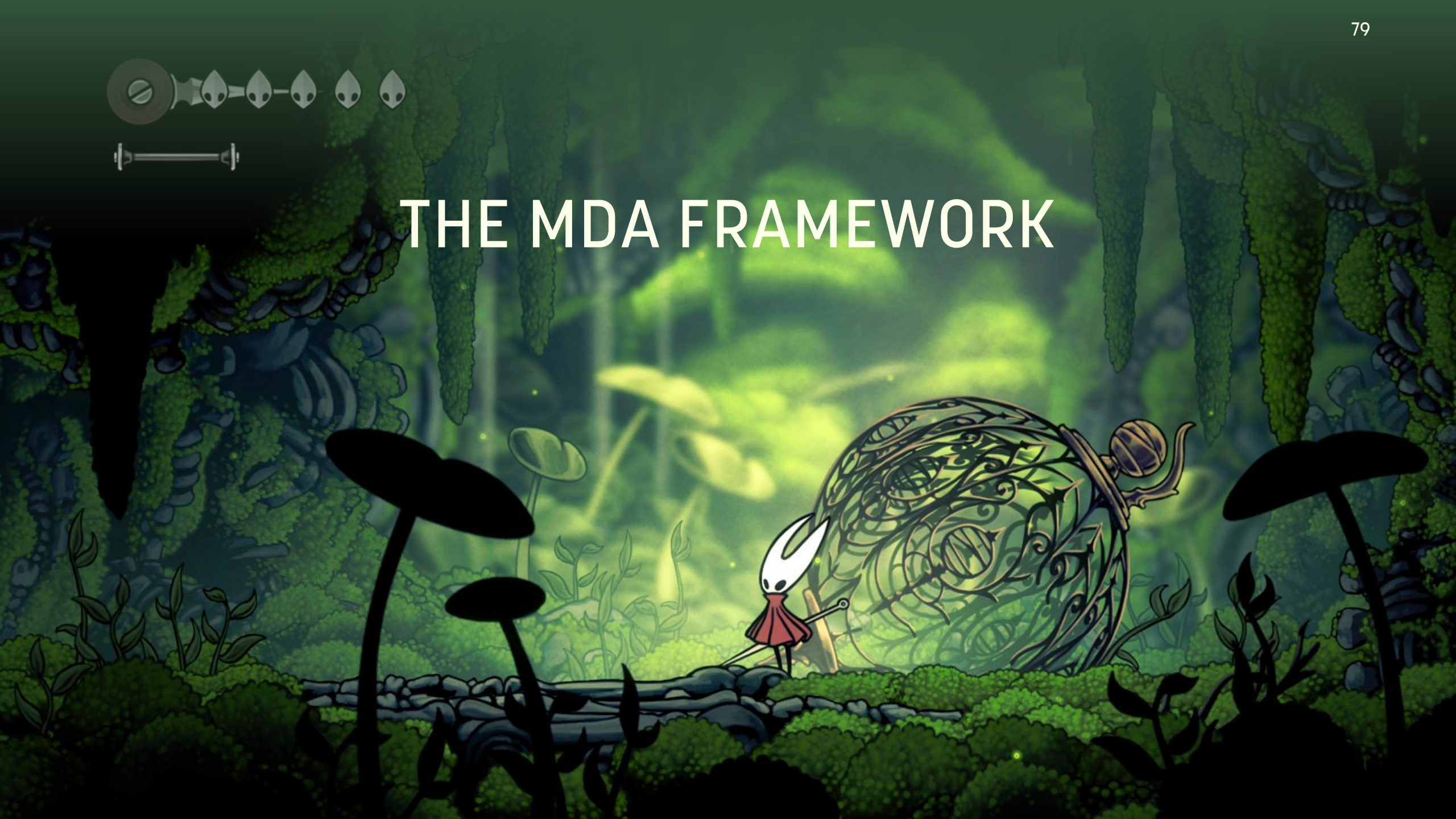
... NOT A CORE MECHANIC

- **Mechanic:** enables players to move items around in their inventories
- Artificial intelligence of NPCs



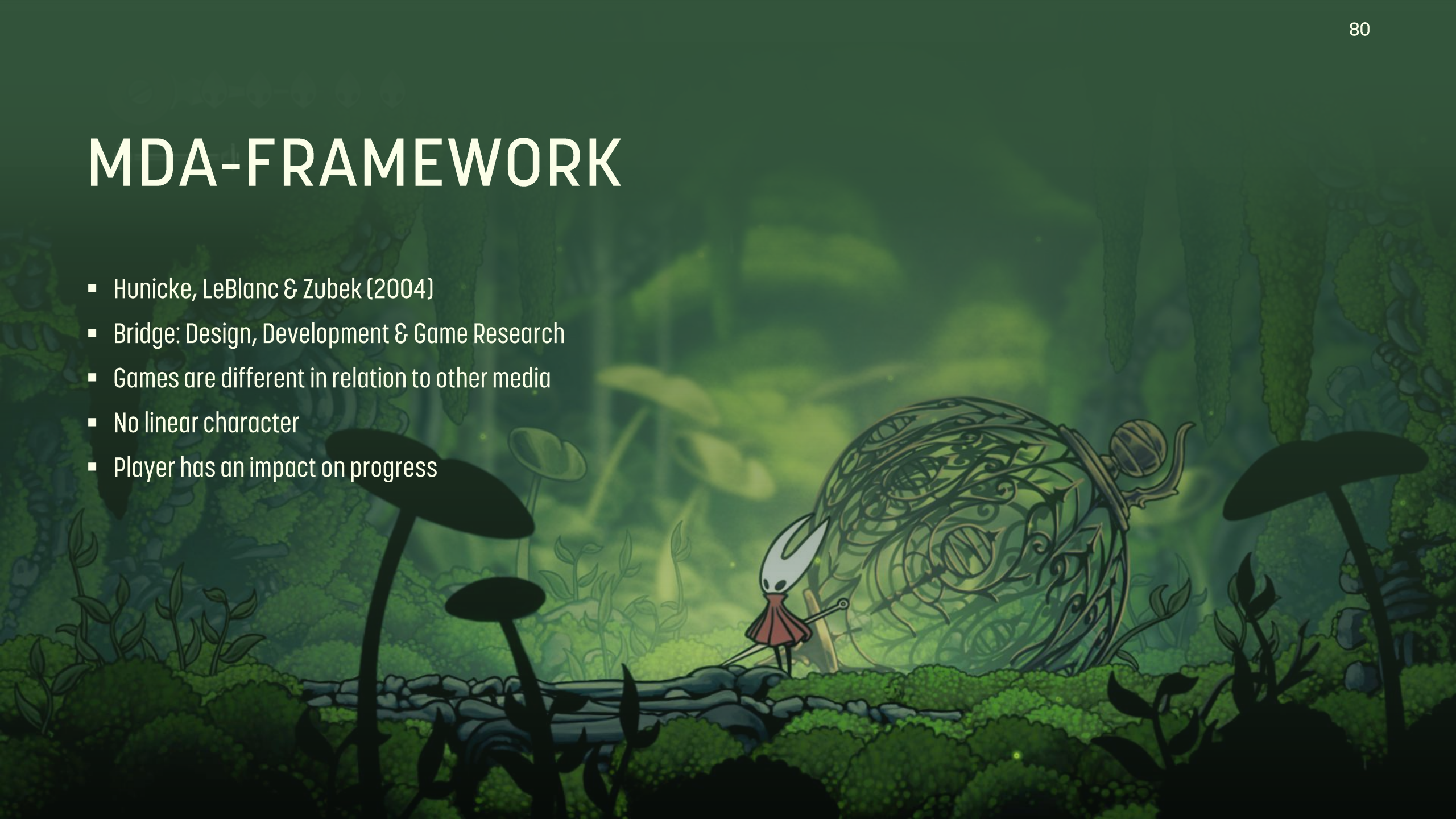


THE MDA FRAMEWORK



MDA-FRAMEWORK

- Hunicke, LeBlanc & Zubek (2004)
- Bridge: Design, Development & Game Research
- Games are different in relation to other media
- No linear character
- Player has an impact on progress





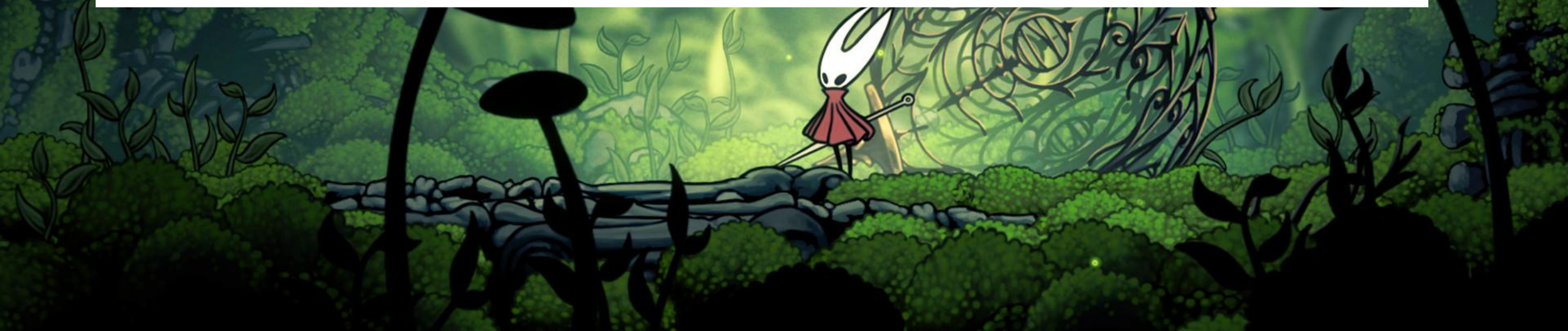
Mechanics



Dynamics



Aesthetics





Rules

System

“Fun”

Mechanics

Dynamics

Aesthetics

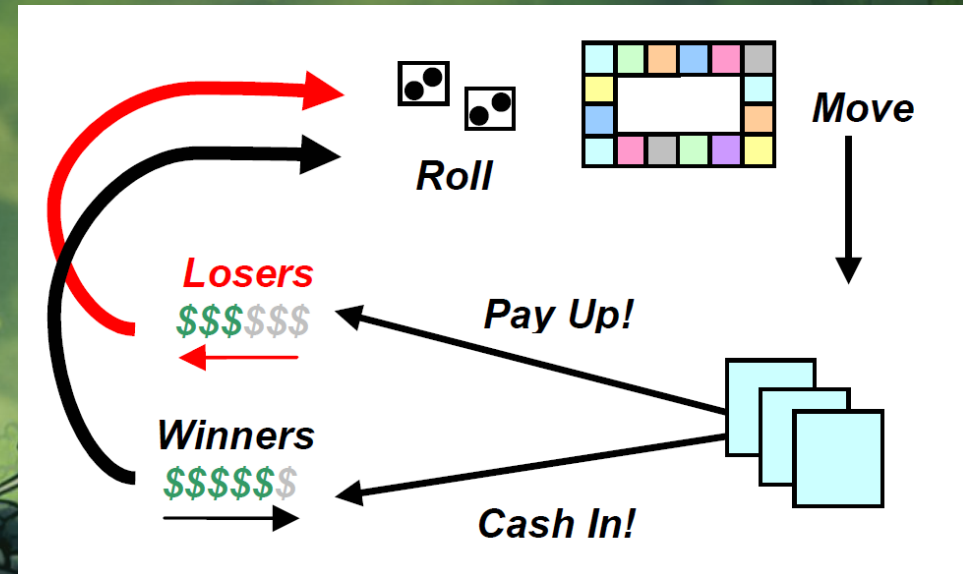
MECHANICS

- **Modify mechanics:** different dynamics
- **Example:** Monopoly
- **Situation:** at a certain time in the game it is hard for players with a low performance to win

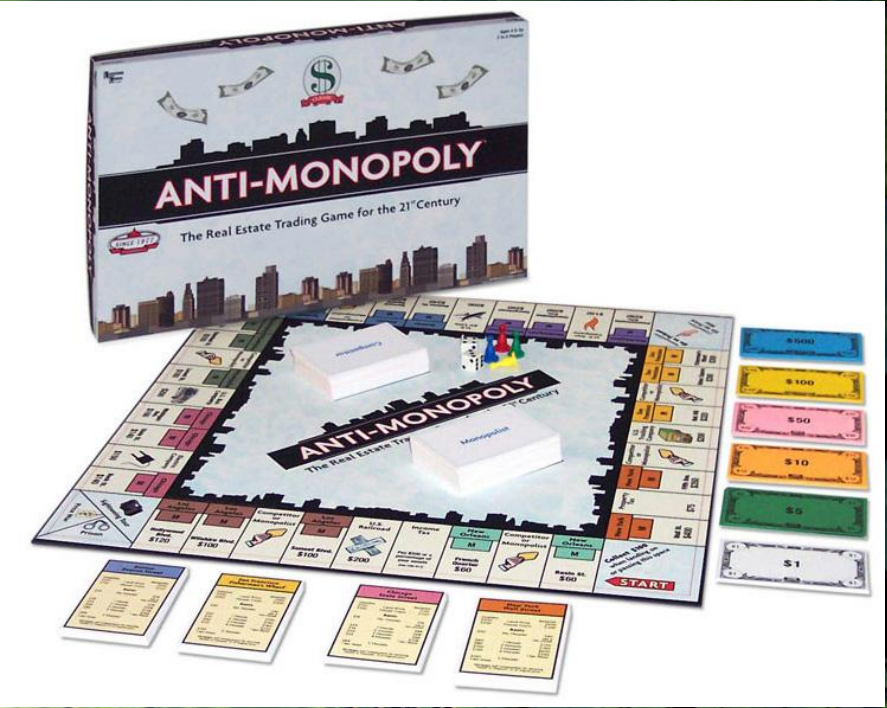


MECHANICS

- Edit game mechanics
- Players in financial trouble could get loans
- Rich players could pay taxes







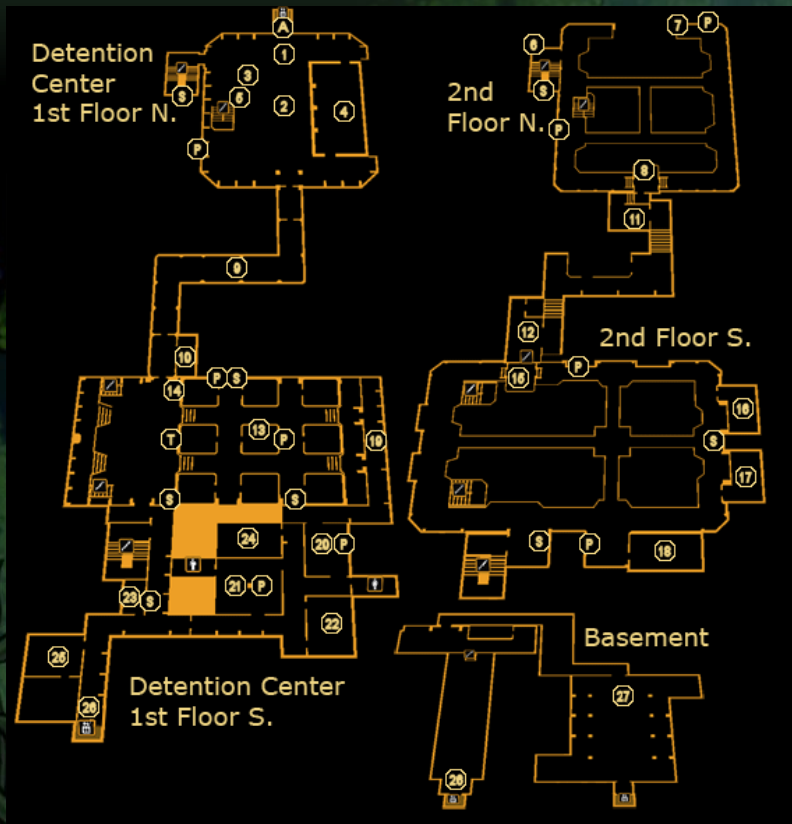
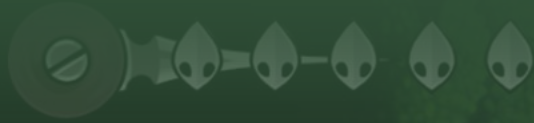
DYNAMICS

- Application of mechanics
- Distinct behavior
- Dynamics influence experience (Aesthetics)
- Example: Deus EX 3



The background of the slide is a screenshot from the game Hollow Knight. It shows the main character, a small white insect-like figure with a red skirt, standing on a mossy ground in a dense, green forest. To the right of the character is a large, ornate, golden structure that looks like a piece of ancient machinery or a large, intricate piece of art. The scene is lit with soft, green light, creating a mysterious atmosphere. In the top left corner, there are faint, semi-transparent icons representing the game's UI, including a compass, a key, and several health or mana icons.

DEUS EX 3: PLAY STYLES











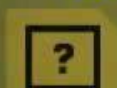


LB X     RB : AUGMENTATIONS

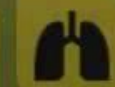
CREDITS 10859 NEXT PRAXIS 3640 TOTAL XP 26360

: CRANIUM AUGMENTATIONS

 0/1
  2/2
  2/2
  0/6

 4/8
  0/2
  1/3
  0/3

: TORSO AUGMENTATIONS

 2/2
  2/7
  0/3
  2/2

: ARM AUGMENTATIONS

 1/8
  0/2



AIM STABILIZER
X VIEW

AVAILABLE PRAXIS 2

: EYE AUGMENTATIONS

 0/1
  1/3

: BACK AUGMENTATIONS

 0/1
  0/1

: SKIN AUGMENTATIONS

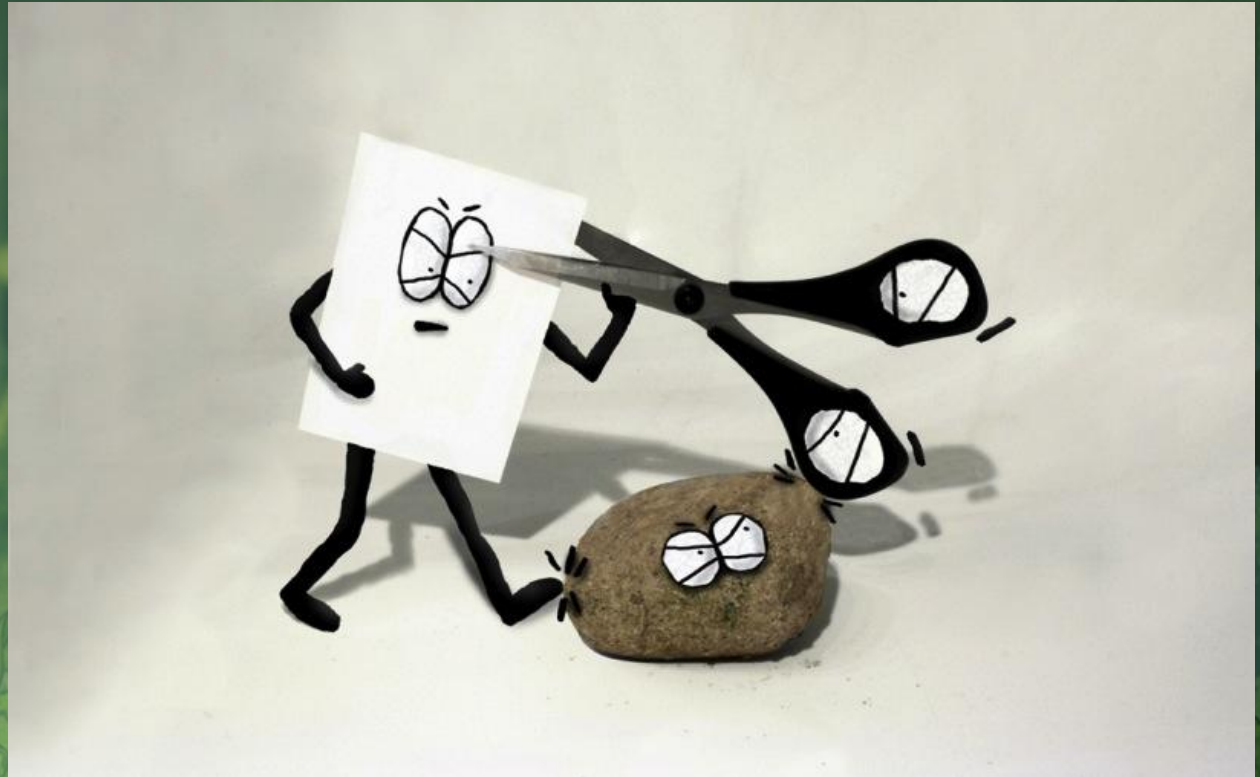
 0/4
  0/3

: LEG AUGMENTATIONS

 0/5

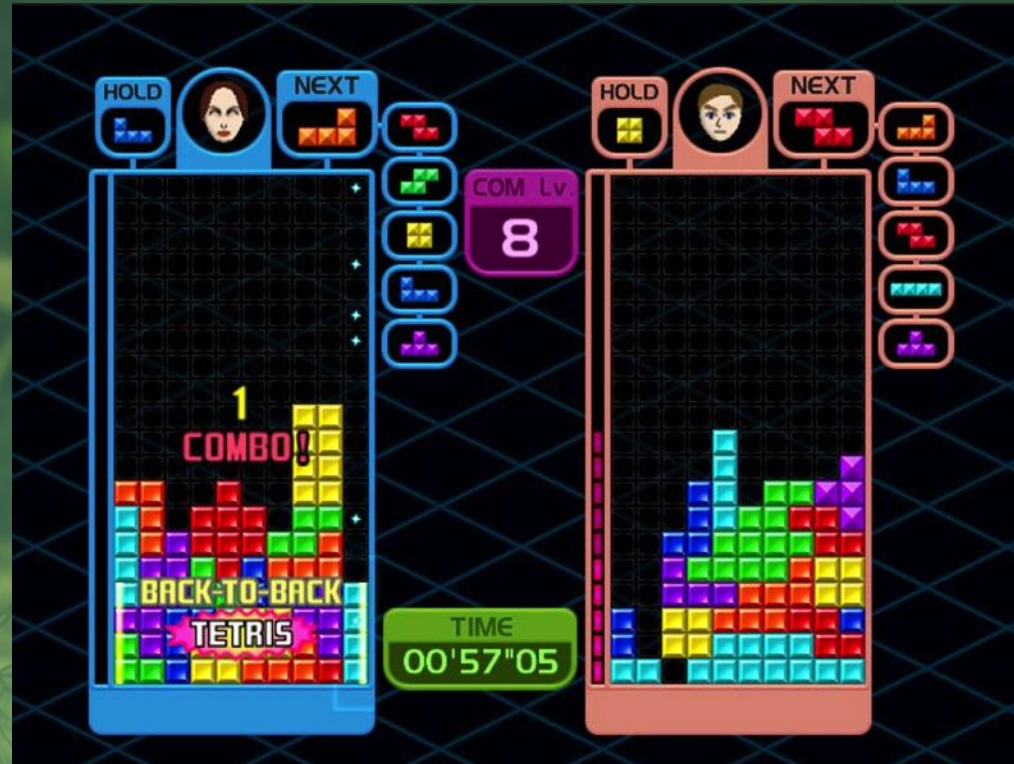
CORE DYNAMICS: PREDICTION

- Examples: roulette
- Do the right thing at the right time
- Use of probability mechanisms



CORE DYNAMICS: SPATIAL REASONING

- Examples: Tetris, Tic-Tac-Toe
- Place elements



CORE DYNAMICS: SURVIVAL

- Example: Resident Evil
- Limited resources



CORE DYNAMICS: DESTRUCTION

- Example: Red Faction
- Often many resources
- Opposite form of survival



CORE DYNAMICS: BUILDING

- RPGs
- Sim City



CORE DYNAMICS: COLLECTION

- Super Mario coins
- Diablo



CORE DYNAMICS: CHASING & EVADING

- Pac Man
- Chase HQ



CORE DYNAMICS: TRADING

- Siedler
- Civilization



CORE DYNAMICS: RACE TO THE END

- Super Mario Kart



24.09.2025

AESTHETICS

- Not only related to the audiovisual layer
- Emotional quality
- Game experience

1. Sensation

Game as sense-pleasure

2. Fantasy

Game as make-believe

3. Narrative

Game as drama

4. Challenge

Game as obstacle course

5. Fellowship

Game as social framework

6. Discovery

Game as uncharted territory

7. Expression

Game as self-discovery

8. Submission

Game as pastime

AESTHETICS



Charades: Fellowship, Expression, Challenge.

Quake: Challenge, Sensation, Competition, Fantasy.

The Sims: Discovery, Fantasy, Expression, Narrative.

Final Fantasy: Fantasy, Narrative, Expression, Discovery, Challenge, Submission.

AESTHETICS

- Designer: often starts with aesthetical considerations
- Game Experience
- Next step: dynamics & mechanics





CLASSIFYING MECHANICS

Five different types



FIVE TYPES OF MECHANICS: PHYSICS

- Define physics in the game world
- Characters commonly move from place to place, jump up and down
- **Cartoon physics** use a modified version of Newtonian mechanics





PHYSICS



The floor is jelly



PHYSICS



<https://www.youtube.com/watch?v=NYSd8gXt060>

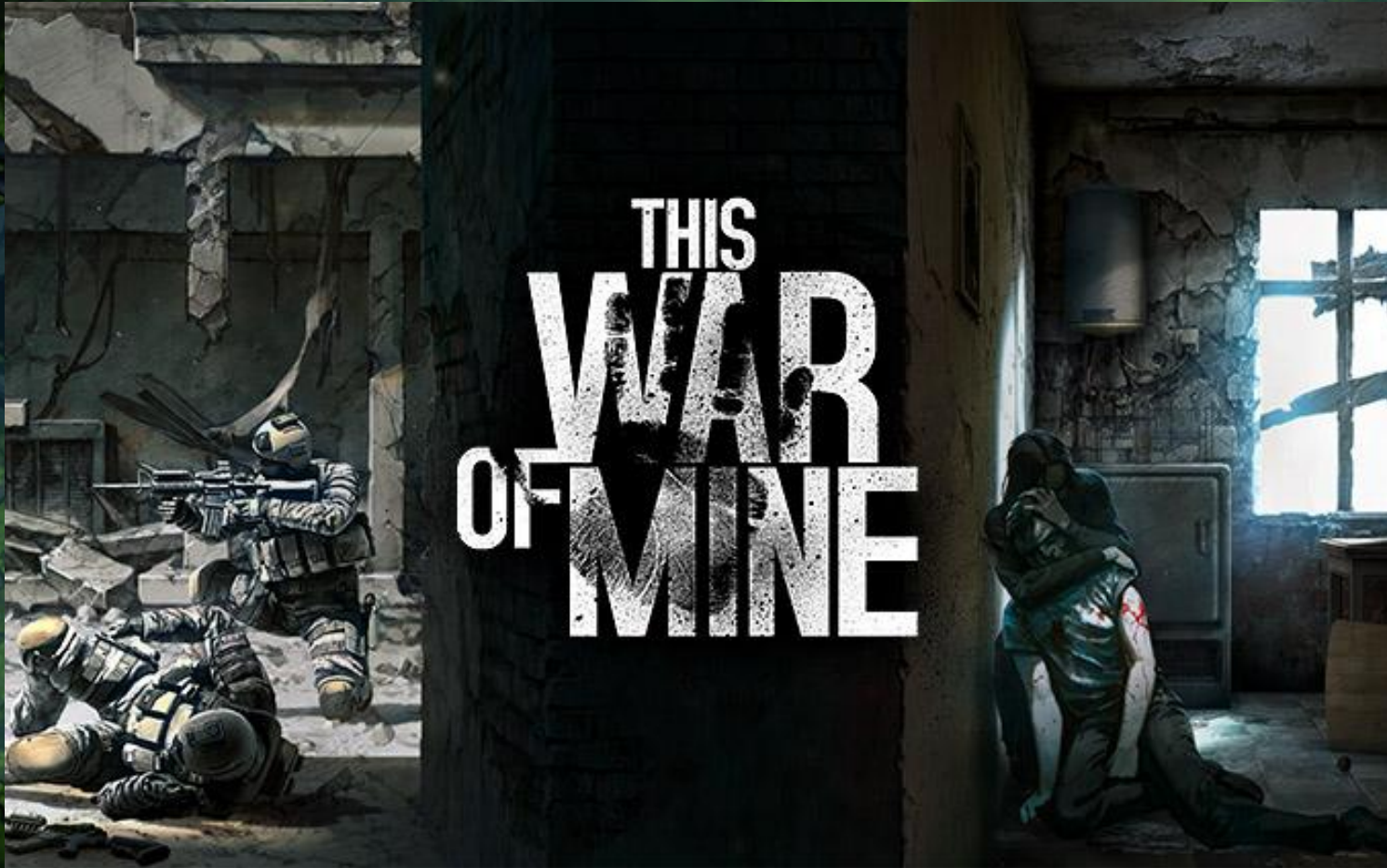
INTERNAL ECONOMY

- Game elements: collected, consumed, and traded
- Easily identified as resources: money, energy, ammunition
- Not limited to concrete, tangible items
- Also include abstractions such as health, popularity, and magic





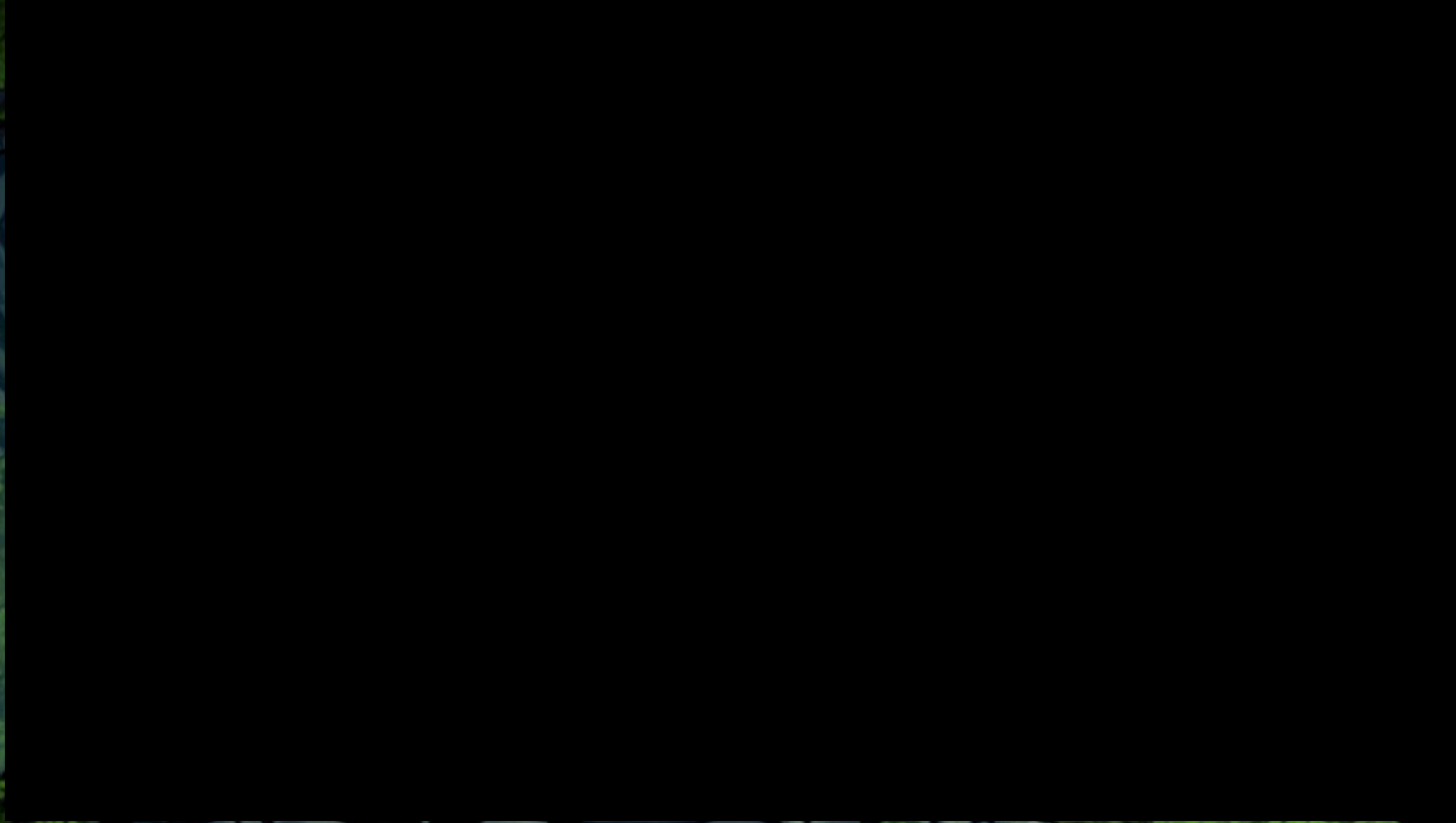
INTERNAL ECONOMY



This war of mine



INTERNAL ECONOMY



<https://www.youtube.com/watch?v=BALBUyoTxQM>

PROGRESSION MECHANISMS

- Often: player's avatar needs to get to a particular place
- Rescue someone or to defeat the main evil
- Complete the level
- Levers, switches, and magical swords





PROGRESSION MECHANISMS

LET'S BEGIN AGAIN.



The Stanley Parable



PROGRESSION MECHANISMS



<https://www.youtube.com/watch?v=fBtXOS2J32Y>

TACTICAL MANEUVERING

- Placement of game units on a map for offensive or defensive advantages
- Mostly seen in strategy games
- Game space: strategic advantages each type of unit
- Many games restrict the location of units to discrete tiles



TACTICAL MANEUVERING



SOCIAL INTERACTION

- Online games
- Mechanics that reward giving gifts
- Inviting new friends to join
- Participating in other social interactions





SOCIAL INTERACTION



Facade

DISCRETE VS. CONTINUOUS MECHANICS

- **Continuous:** simulate physics with precise mechanics that create a smooth flow of play
- **Game object:** half a pixel more to the left...
... can have a huge effect on the result of a jump
- **Twitch-Skill**



DISCRETE VS. CONTINUOUS MECHANICS

- Discrete
- Internal economy tend to be discrete
- represented with integer values
- Game elements and actions: finite set
- No gradual transitions
- Strategic skill



MECHANICS AND GAME GENRES

	Physics	Economy	Progression	Tactical Maneuvering	Social Interaction
Action	Detailed physics for movement, shooting, jumping, etc.	Power-ups, collectables, points and lives	Predesigned levels with increasingly difficult tasks, storyline to set player goals		
Strategy	Simple physics for movement and fighting	Unit building, resource harvesting, unit upgrading, risking units in combat	Scenarios to provide new sets of challenges	Positioning of units to gain offensive or defensive advantages	Coordinated actions, alliances and competition between players
Role-Playing	Relatively simple physics to resolve movement and conflict, often turn-based	Equipment and experience to customize a character or party	Story line and quests to give player a purpose and goal	Party tactics	Play-acting
Sports	Detailed simulation	Team management	Seasons, competitions, tournaments	Team tactics	

MECHANICS AND GAME GENRES

Vehicle Simulation	Detailed simulation	Vehicle tuning between missions	Missions, races, challenges, competitions, tournaments	
Management Simulation		Managing of resources, economy building	Scenarios to provide new sets of challenges	Managing of resources, economy building
Adventure		Managing a player's inventory	Story to drive game, locks and key to control player progress	Coordinated actions, alliances and competition between players
Puzzle	Simple, often non-realistic and discrete, physics generate challenges		Short levels providing increasingly more difficult challenges	
Social Games		Resource harvesting and unit building, resources spend on personalized content	Quests and challenges to give player a purpose and a goal	Players exchange in-game resources, mechanics encourage player cooperation or conflict

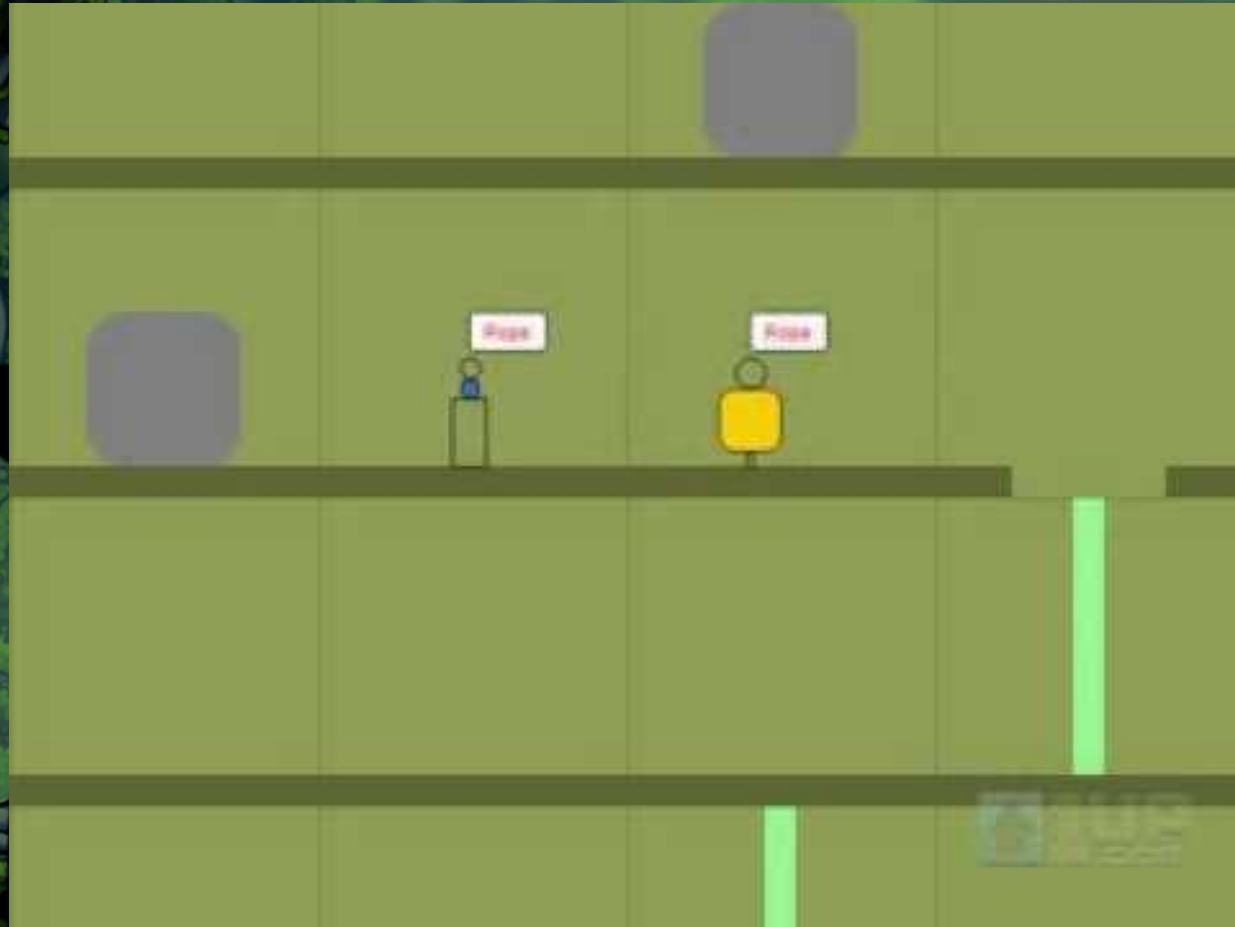


PRACTICAL IMPLICATIONS?





GAME MECHANICS WITH TWITCH SKILLS





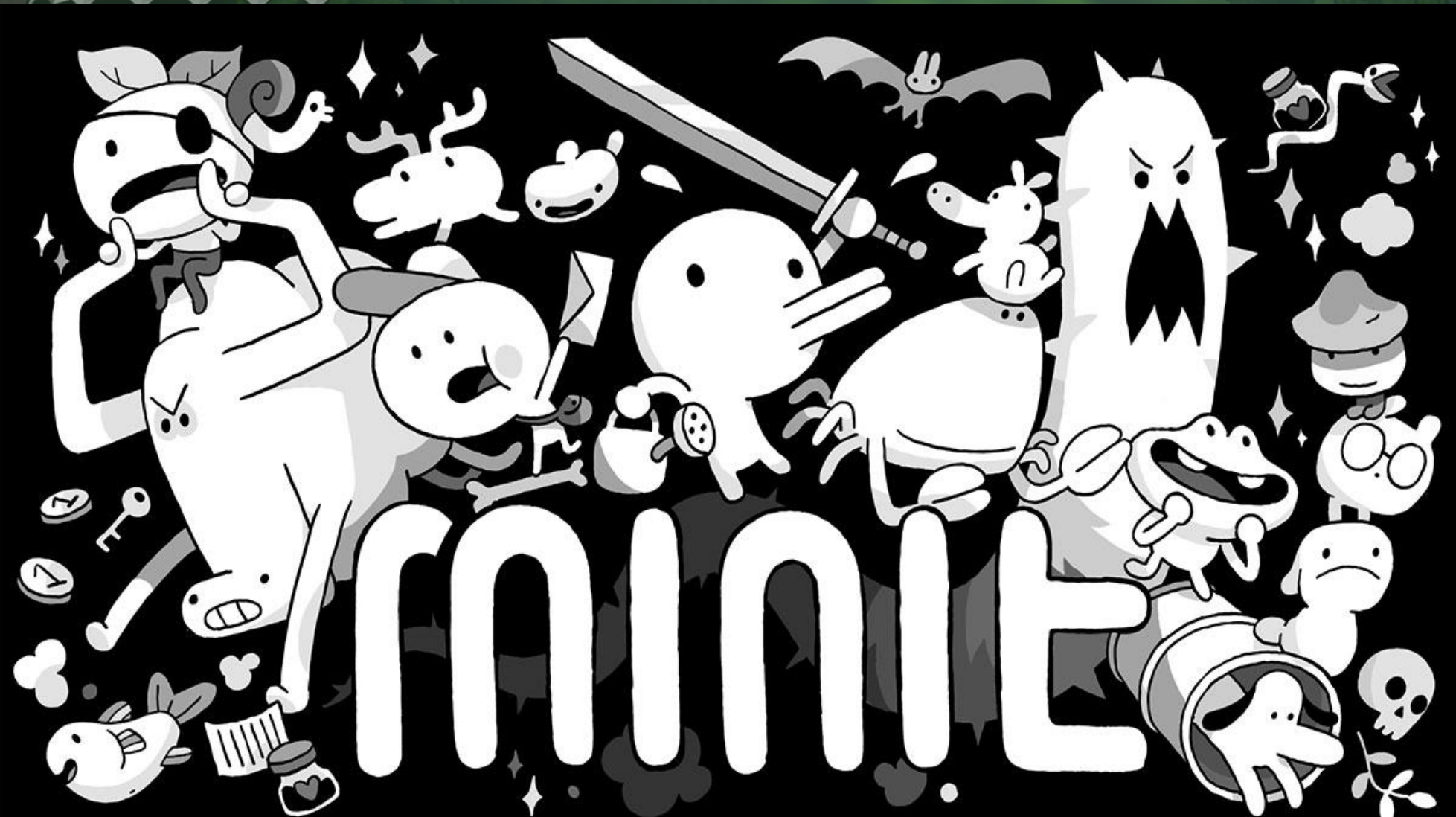
GAME MECHANICS WITH TWITCH SKILLS





The image features a dark green, atmospheric background from the game Hollow Knight. In the upper left corner, there is a semi-transparent overlay of the game's UI, including a health bar with six heart icons (the first is partially filled) and a long, thin spirit meter below it. The main scene depicts the character Hollow Knight, a small white insect-like figure with a red skirt, standing on a mossy ledge. The knight is looking towards a large, ornate, spherical structure made of intricate, glowing green patterns, which is part of the game's environment. The scene is dimly lit with soft green light filtering through the background, creating a mysterious and somber mood.

GAME RECOMMENDATION





mini

The background is a dark, atmospheric illustration of a forest from the game Hollow Knight. A small, white, insect-like character with a red skirt and a crescent-shaped head is standing on a mossy ground. The character is holding a long, thin, golden object, possibly a key or a staff, and is looking towards a large, ornate, golden structure that resembles a nest or a piece of ancient machinery. The forest is filled with large, dark mushrooms, hanging vines, and various plants. The lighting is dim, with some light filtering through the trees, creating a sense of mystery and depth. In the top left corner, there are faint, semi-transparent icons of various weapons and abilities, and a small horizontal bar with a key icon, likely representing the player's inventory or a specific game mechanic.

THANK YOU FOR YOUR ATTENTION!